

Information Asymmetry, Mispricing, and Security Issuance

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ABSTRACT

I examine the effects of information asymmetry-driven mispricing on security issuance. Using predisclosure changes in purchase obligations as a proxy for information asymmetry-driven mispricing, I find that managers avoid (prefer) issuing securities when they perceive their firms to be undervalued (overvalued). The effects of information asymmetry-driven mispricing are stronger on equity issuance than debt issuance. Consequently, undervaluation (overvaluation) causes an increase (decrease) in leverage. These effects are more pronounced for firms, periods, and securities associated with greater information asymmetry. The stock-trading patterns that managers follow suggest that their perceived mispricing is an important factor in both private and firm-level decisions.

A MANAGER TYPICALLY HAS SUPERIOR information about his or her firm's future performance relative to what outside investors know. This information asymmetry can cause a firm to be undervalued or overvalued in the market relative to the manager's perception. Such information asymmetry-driven mispricing may affect a range of managerial decisions. In this paper, I investigate the effect of information asymmetry-driven mispricing on security issuance.

Addressing this question presents a challenge because it is difficult to measure mispricing caused by information asymmetry. First, measuring mispricing, regardless of the cause, is difficult because the true value of firm shares is not observable. Some studies on market timing use the market-to-book ratio as a proxy for mispricing (Baker and Wurgler (2002), DeAngelo, DeAngelo, and Stulz (2010)). A key drawback of this method is that the market-to-book ratio can also proxy for growth options or time-varying adverse selection costs. Other studies interpret preissuance share price run-ups and postissuance

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underperformance as evidence of market timing (Loughran and Ritter (1995, 1997), DeAngelo, DeAngelo, and Stulz (2010)). However, these findings can be explained by Q-theory (Li, Livdan, and Zhang (2009)) or rational theory in a real options framework (Carlson, Fisher, and Giammarino (2006)). Several studies use an earnings-based valuation model to estimate the mispricing of a share relative to its economic value (D'Mello and Shroff (2000), Rhodes-Kropf, Robinson, and Viswanathan (2005), Elliott, Koeter-Kant, and Warr (2008)), but these studies assume that managers forecast future earnings perfectly, which is unrealistic. More importantly, the methods above do not allow us to determine whether the mispricing is caused by investor irrationality or information asymmetry.

To overcome this challenge, I exploit the mandatory disclosure of purchase obligations (POs) in 10-K filings.¹ The U.S. Securities and Exchange Commission (SEC) defines a PO as an enforceable and legally binding agreement to purchase goods or services in the future. A PO thus represents a quantity of production inputs that a firm has committed to purchase in the near future. A PO likely increases (decreases) when a manager expects demand for the firm's products to increase (decrease). A PO can therefore proxy for a manager's expectations regarding the firm's future performance. Furthermore, although POs are made throughout the year, only snapshots of POs are disclosed in 10-Ks at the end of each year, that is, outside investors do not receive information on POs until they are disclosed in 10-K filings. As such, POs proxy for a manager's forecasts of the firm's future performance but remain unknown to investors over the course of a given year. In this paper, I exploit predisclosure changes in the POs of manufacturing firms as a proxy for mispricing relative to a manager's private information about firm value to examine the effect of information asymmetry-driven mispricing on security issuance.

Two conditions need to be satisfied to validate the use of predisclosure changes in POs as a proxy for information asymmetry-driven mispricing. First, predisclosure changes in POs should predict future changes in security prices. If predisclosure changes in POs are not positively related to future changes in security prices, the assumption that POs represent a manager's superior information about firm value cannot be satisfied. Second, predisclosure changes in POs should be private information. If outside investors have the same updated information as managers throughout the year, predisclosure changes in POs would not predict mispricing.

To verify the first condition, I examine whether changes in POs predict changes in future stock prices. To this end, I define ΔPO_t as PO_t minus PO_{t-1} , where PO_t is the total dollar amount of POs recorded in 10-Ks in fiscal year t scaled by total assets. Similarly, I define ΔPO_t^{t+1} as PO_t^{t+1} minus PO_{t-1}^t , where PO_t^{t+1} is the dollar amount of POs recorded in 10-Ks in fiscal year t that are due during year $t+1$ scaled by total assets, and I define ΔPO_t^{rest} as PO_t^{rest} minus PO_{t-1}^{rest} , where PO_t^{rest} is PO_t minus PO_t^{t+1} . I find that ΔPO_t (ΔPO_t^{t+1}) is

¹ Companies are not required to disclose POs in their quarterly reports (typically on Form 10-Q) unless there are material changes outside the ordinary course of business during a given quarter.

positively associated with size and book-to-market (BM)-adjusted buy-and-hold returns during $t+1$ ($BHAR_{t+1}$), which are computed by adding daily abnormal returns for all trading days during fiscal year $t+1$. In particular, a one-standard-deviation increase in ΔPO_t (ΔPO_t^{t+1}) is associated with an increase in $BHAR_{t+1}$ of approximately 1.8 (2.7) percentage points, which accounts for approximately 5.1% (7.7%) of the standard deviation of $BHAR_{t+1}$. The estimate also implies that an increase in ΔPO_t (ΔPO_t^{t+1}) from the 25th percentile to the 75th percentile increases $BHAR_{t+1}$ by 0.3 (0.6) percentage points.²

To verify the second condition that prediscovered POs constitute private information I investigate stock price responses following the disclosure of POs. If prediscovered POs constitute private information, stock prices should respond immediately in an appropriate direction following the disclosure of 10-Ks. The results confirm this expectation. A one-standard-deviation increase in ΔPO_t (ΔPO_t^{t+1}) is associated with an increase in $BHAR(-1,1)$ of approximately 28 (33) bps, which accounts for 7.6% (8.9%) of its standard deviation.³ The estimate also implies that an increase in ΔPO_t (ΔPO_t^{t+1}) from the 25th percentile to the 75th percentile increases $BHAR(-1,1)$ by 4 (8) bps. Taken together, these two sets of results support the use of prediscovered changes in POs as a proxy for information asymmetry-driven mispricing.⁴

I next turn to the effects of information asymmetry-driven mispricing on securities issuance. If a manager has private information that the firm's value will increase in the near future, that is, that the firm is currently undervalued, then the manager is likely to avoid issuing securities and may even pursue stock repurchases or debt retirements.⁵ Similarly, if a manager's private information indicates that the firm's value will decrease in the future, that is, that the firm is currently overvalued, then their incentives to issue securities strengthen.⁶ If the value of debt is less sensitive to information than that of equity, however, the effects of information asymmetry-driven mispricing on net debt issues would be smaller than those on net equity issues. Consequently, a firm's financial leverage would increase (decrease) when it is undervalued (overvalued) based on a manager's perception. Because increases (decreases) in POs indicate undervaluation (overvaluation), changes in POs are expected to be negatively associated with contemporaneous net security issuance, especially net equity issuance, but positively associated with financial leverage.

² The differences in the economic magnitudes of the standard deviation- and interquartile-based interpretations arise from the skewness of ΔPO_t . For example, the standard deviation of ΔPO_t is 0.062, while the interquartile range of ΔPO_t is 0.012.

³ In particular, $BHAR(-1,1)$ is size and BM-adjusted buy-and-hold returns during a three-day period starting one day before the 10-K filing date.

⁴ The implicit assumption is that investors treat POs as a random walk.

⁵ Graham and Harvey's (2001) survey evidence reveals that firms are reluctant to issue common stock when they perceive it to be undervalued.

⁶ The assumption is that a manager cares about existing shareholders.

My empirical results support this logic. Controlling for firm characteristics, firm fixed effects and industry-year fixed effects,⁷ a one-standard-deviation decrease in ΔPO_t (ΔPO_t^{t+1}) is associated with an increase in contemporaneous net equity issuance of 1.1% (1.8%) of total securities outstanding, which accounts for approximately 8.7% (13.8%) of the sample standard deviation of net equity issuance. The estimate also implies that a decrease in ΔPO_t (ΔPO_t^{t+1}) from the 75th percentile to the 25th percentile increases equity issuance by 0.2% (0.4%) of total securities outstanding. On the other hand, a one-standard-deviation decrease in ΔPO_t (ΔPO_t^{t+1}) is associated with an increase in contemporaneous net debt issuance of 0.23% (0.35%) of total securities outstanding, which accounts for approximately 2.3% (3.5%) of the sample standard deviation of net debt issuance. Another economic implication of this result is that a decrease in ΔPO_t (ΔPO_t^{t+1}) from the 75th percentile to the 25th percentile is expected to increase debt issuance by 0.03% (0.08%) of total securities outstanding. Thus, the effect of ΔPO_t on net equity issuance is significantly stronger than that on net debt issuance, which implies that ΔPO_t is positively associated with contemporaneous financial leverage.

I provide empirical evidence that supports the interpretation that the negative association between ΔPO_t and security issuance is due to information asymmetry. I first exploit cross-sectional variation in the degree of information asymmetry and show that the negative association between ΔPO_t and issuance is stronger among firms with more severe information asymmetry problems. For example, I do not find a significant association between ΔPO_t and contemporaneous security issuance among firms that regularly report POs on their 10-Qs, which are expected to suffer less severe information asymmetry about POs compared with firms that report POs only on annual reports. In addition, using frequently used proxies for the level of information asymmetry, such as analyst coverage, firm age, and R&D intensity, I find the negative association between ΔPO_t and contemporaneous security issuance to be stronger among high-information-asymmetry firms. Next, I exploit time-varying information asymmetry. I find that the negative association between ΔPO_t and quarterly net security issuance is strongest in the last quarter of year t , when information asymmetry about ΔPO_t is expected to be most severe. In addition, I find no association between ΔPO_t and net security issuance during the first quarter of $t+1$, indicating that the negative association disappears following disclosure of POs. Furthermore, I exploit variation in the degree of information sensitivity between public debt and private debt. I find that the negative association between ΔPO_t and net debt issuance is due to public debt, which is more information sensitive.

A potential concern with the analysis above is that the negative association between ΔPO_t and contemporaneous security issuance is due to time-varying fundamentals that are correlated with POs. For example, a decrease (increase)

⁷ I also control for contemporaneous stock returns and one-year-lagged stock returns in all regressions. The coefficient estimates on ΔPO_t (ΔPO_t^{t+1}) therefore capture information over and above what is already incorporated in stock prices.

in POs may be associated with increased (decreased) demand for external financing. However, I do not find any association between ΔPO_t and contemporaneous or future financing deficit.

Lastly, managers' stock trading and option exercise patterns are consistent with the view that they believe changes in POs predict mispricing. I find that managers engage in significantly more stock sales (purchases) when predis-closed POs predict overvaluation (undervaluation) of shares. Similarly, managers' option exercises are negatively associated with ΔPO_t . These results suggest that managers' perceived mispricing is an important determinant of both private and firm-level decisions.

In addition to contributing to the literature on market timing, as explained above, this study contributes to studies on information asymmetry and capital structure. The pecking-order theory of Myers (1984) and Myers and Majluf (1984) implies that firms should prefer debt to equity because equity involves a higher level of information asymmetry than debt. Consistent with this prediction, Bharath, Pasquariello, and Wu (2009) find that firms with greater information asymmetry fund a greater percentage of their financing deficit with debt. Although both Bharath, Pasquariello, and Wu (2009) and my study examine information asymmetry and security issuance, Bharath, Pasquariello, and Wu's (2009) measure of information asymmetry is based on the assessment of adverse-selection risk in the secondary market between informed and uninformed investors, whereas my measure is based directly on the private information held by managers and thus identifies information asymmetry between insiders and outsiders more cleanly. More importantly, Bharath, Pasquariello, and Wu (2009) do not consider the sign of mispricing, that is, whether information asymmetry causes underpricing or overpricing of firm shares, while my findings indicate that the sign of mispricing plays an important role in capital structure.

Korajczyk, Lucas, and McDonald (1991, 1992) consider time-varying information asymmetry. Korajczyk, Lucas, and McDonald (1991) use earnings announcements as information events and show that firms tend to issue equity following earnings announcements, especially those with positive stock price responses.⁸ More importantly, they show that the price drop on the date of an equity issuance announcement increases since the last information release. Kerr and Ozel (2015) extend Korajczyk, Lucas, and McDonald (1991) by analyzing both debt and equity issuance and find that clustering of issuance after earnings announcements increases with a security's sensitivity to information asymmetry. Unlike Korajczyk, Lucas, and McDonald (1991), however, Kerr and Ozel (2015) find that the fraction of bond offerings and seasoned equity offerings (SEOs) made after earnings announcements that reveal good news is

⁸ The results reported in columns (5) and (6) of both panels of Table VII indicate that there is no association between ΔPO_{t-1} and security issuance during the first quarter of year t , which suggests that my findings are not driven by the mechanism proposed by Korajczyk, Lucas, and McDonald (1991), namely, that managers issue securities following releases of favorable information about the firm.

largely statistically indistinguishable from those that reveal bad news. They attribute this result to firms' desire to avoid the potential costs associated with the omission of negative material information. Although these studies and my paper investigate time-varying information asymmetry and security issues, there is an important difference. In particular, these studies do not directly measure managers' private information but instead show evidence consistent with information asymmetry increasing since the previous information releases, whereas I measure managers' private information based on POs and thus am able to provide more direct evidence on the implications of information asymmetry.

This paper is also related to Lee (2010), who empirically shows that, during the 2004 to 2007 period, firms entered into additional POs when they expected demand for their projects to increase. In particular, he finds that growth in POs is positively associated with future sales, earnings, and abnormal stock returns. Unlike what I find, however, he finds that the market fails to fully incorporate the implications of POs, as announcement returns are not significantly associated with changes in POs, resulting in longer delays in stock price responses than I observe in my sample firms.⁹

This paper also engages with the insider trading literature. A prevalent finding of this literature is that insiders use the private information they possess at the time they trade. Several studies identify particular types of private information that managers use, including foreknowledge of future earnings (Ke, Huddart, and Petroni (2003)), disclosure of material corporate events such as bankruptcies (Seyhun and Bradley (1997)), dividend initiations (John and Lang (1991)), SEOs (Karpoff and Lee (1991)), and stock repurchases (Lee, Mikkelsen, and Partch (1992)). I contribute to this literature by adding the information reflected in POs as a specific type of private information that managers use to their advantage. Compared with trades executed prior to earnings announcements or disclosures of material corporate events, trades based on predisclosed POs may be less subject to legal jeopardy and as a result could be more prevalent. This paper is also related to Jenter (2005), who shows that managers' private trading and firm-level security issuance are closely related. My study differs from Jenter (2005) as I link private trades and firm-level security issues through a particular source of private information that managers possess. In contrast, Jenter (2005) suggests that insiders may not use valid private information.

The remainder of this paper proceeds as follows. In Section I, I introduce the baseline empirical strategy and describe the data. I validate the use of predisclosure changes in POs as a proxy for information asymmetry-driven mispricing in Section II. In Section III, I present results of the baseline tests. Section IV presents results of tests that exploit variation in the levels of information asymmetry. I address alternative explanations in Section V. Section VI

⁹ In the [Internet Appendix](#), I report the results obtained by replicating Lee's (2010) procedure. The [Internet Appendix](#) may be found in the online version of this article.

presents evidence that managers believe changes in POs predict mispricing of shares. I conclude in Section VII.

I. Baseline Strategy and Data

A. POs and Baseline Strategy

In January 2003, the SEC issued a final rule to implement Section 401(a) of the Sarbanes-Oxley Act of 2002, which requires public firms to disclose overviews of their off-balance-sheet arrangements in their 10-Ks for fiscal years ending on or after December 15, 2003. The rule requires companies to disclose an overview of contractual obligations in tabular format in their annual reports (typically Form 10-K). POs are among the four categories of contractual obligations affected, along with long-term debt obligations, capital lease obligations, and operating lease obligations. The SEC defines a PO as an enforceable and legally binding agreement to purchase goods or services in the future and requires firms to disclose both the total amount of POs and the approximate timing of payments due in future periods. POs thus indicate the quantity of production inputs a firm has committed to purchase.

POs likely increase (decrease) when a manager expects demand for the firm's products to increase (decrease). POs can therefore be regarded as reflecting the manager's expectations about the firm's future performance. Furthermore, while POs are made throughout a year, only snapshots of them are disclosed in 10-Ks at each year-end and thus outside investors do not have this information during the fiscal year. As such, POs proxy for a manager's forecasts of a firm's future performance but remain unknown to investors throughout a given year.

My empirical strategy is to use predisclosed changes in POs as a proxy for information asymmetry-driven mispricing to examine the effects of mispricing on security issuance and capital structure. An increase in POs would indicate that a manager holds private information suggesting that the performance and value of the firm will increase, and thus that the firm is undervalued in the current market. In contrast, a decrease in POs would indicate that the manager holds private information suggesting that the firm's future performance and value will decline, and thus that the firm is overvalued in the current market. For this strategy to work, however, two conditions must be satisfied. First, predisclosure changes in POs should predict future changes in security prices. Second, outside investors should not have the information included in such POs before disclosure. Below I begin by ensuring that these two conditions are satisfied. I then examine the effects of changes in POs on contemporaneous security issuance and financial leverage.

B. Data and Variables

I hand-collect information on POs from 10-K filings for the period 2003 to 2015. I consider a sample of public manufacturing firms (SICs 2000 to 3999) in the Compustat universe. I separately record information on the total dollar

amount of POs and POs due in the following year. If a firm does not report POs, I record them as zero. I exclude firms that never report POs during the sample period. Other financial data come from Compustat and stock return data from the Center for Research in Security Prices (CRSP). The sample of seasoned common equity offerings and stock repurchases is obtained from Thomson Financial's Securities Data Corporation (SDC) database. Analysts' earnings forecasts, actual realized earnings, and analyst coverage are taken from Institutional Brokers' Estimates System International, Inc. (I/B/E/S). Data on managers' trading activities are collected from Form 4 reported to the SEC, and data on public debt are obtained from Capital IQ. Lastly, data on managers' option exercises and holdings come from Execucomp.

The main variables of interest are *NetEquityIssue* and *NetDebtIssue*. I define *NetEquityIssue* and *NetDebtIssue* as the difference between issues and repurchases or retirements scaled by lagged total securities outstanding, which is the sum of debt and equity, as shown in equations (1) and (2):

$$NetEquityIssue_t = \frac{EquityIssues_t - EquityRepurchases_t}{Debt_{t-1} + Equity_{t-1}}, \quad (1)$$

$$NetDebtIssue_t = \frac{DebtIssues_t - DebtRetirements_t}{Debt_{t-1} + Equity_{t-1}}. \quad (2)$$

I define *Debt* as total debt (Compustat item DLC plus item DLTT), and *Equity* as the market value of shareholders' equity (*CSHO* × *PRCC_F*). Following a number of studies, I define *EquityIssues* as the total amount of funds received from issuance of common and preferred stock (Compustat item SSTK), and *EquityRepurchases* as the total amount of funds used to repurchase common and preferred stock (PRSTKC) (Baker and Wurgler (2002), Chang, Dasgupta, and Hillary (2006), Kisgen (2006), Butler et al. (2011), Malmendier, Tate, and Yan (2011), Kahle and Stulz (2013), Balakrishnan et al. (2014), Leary and Roberts (2014)). I define *DebtIssues* as long-term debt issuance (DLTIS), and *DebtRetirements* as long-term debt reduction (DLTR).

I define *EquityIssueD_t* as an indicator variable that is equal to one if a firm issues equity valued at more than 1% of the lagged value of total securities, and zero otherwise. Similarly, I define *RepurchaseD_t* as an indicator variable that is equal to one if a firm repurchases equity valued at more than 1% of the lagged value of total securities, and zero otherwise. I define *DebtIssueD_t* as an indicator variable equal to one if a firm issues debt valued at more than 1% of the lagged value of total securities, and zero otherwise. Similarly, I define *DebtReductionD_t* as an indicator variable equal to one if a firm reduces debt valued at more than 1% of the lagged value of total securities, and zero otherwise.

Following Welch (2011), to remove the effects of nonfinancial liabilities, I define leverage (*Leverage*) as debt over the sum of debt and the book value of equity. I define investment (*Investment*) as capital expenditures, scaled by

lagged property, plant, and equipment. Tobin's Q (*Tobin'sQ*) is defined as total assets minus common equity plus the market value of equity, scaled by total assets. Firm size (*Size*) is the natural logarithm of total assets and cash flow (*Cashflow*) is earnings before interest, taxes, depreciation, and amortization (EBITDA) scaled by lagged total assets. Cash holdings (*Cash*) are cash and cash equivalents, scaled by total assets, and tangibility (*Tangibility*) is defined as property, plant, and equipment, scaled by total assets. I define $BHAR(-1,1)$ as size and BM-adjusted buy-and-hold returns during a three-day period starting one day before the 10-K filing date. In particular, I use size and BM-matched portfolios as benchmarks to compute abnormal stock price performance. In constructing benchmark portfolios, I first sort firms into quintiles based on the 20th, 40th, 60th, and 80th percentiles of the market value of shareholders' equity. I then sort each size quintile portfolio into five equal BM quintiles. I define $BHAR_t$ as size and BM-adjusted buy-and-hold returns during fiscal year t . In particular, I add daily abnormal returns of all trading days during t . All other variables are defined in Appendix A and winsorized at 1% in both tails of the distribution.

Table I presents summary statistics. Approximately 95% of the firm-year observations show positive POs outstanding. On average, sample firms have POs of approximately 9.0% of total assets, most of which are due in the following year. An average *NetEquityIssue* is 0.017 and *NetDebtIssue* is 0.011. In addition, 13.5% of sample firms issue equity, while 18.3% repurchase equity, 20.3% issue debt, and 19.1% reduce debt. Sample firms' average *Leverage* is 0.283, and on average sample firms make investments of approximately 33.9% of the value of property, plant, and equipment.

Regarding variables that capture managers' trading activities, I define $NetSell_t$ as the number of shares sold minus the number of shares purchased, scaled by the lagged total number of shares outstanding, a definition that is similar to that in Sias and Whidbee (2010).¹⁰ Because the range of the variable resulting from this scaling is a small interval around zero, I multiply the variable by 1,000. The average of $NetSell_t$ is 0.234, indicating that transactions executed by managers are predominantly sales. I define $Sell_t$ as the number of shares sold times 1,000, and Buy_t as the number of shares purchased times 1,000, both scaled by the lagged total number of shares outstanding. In addition, I define $Sell/(Sell + Buy)_t$ as the number of shares sold over the sum of the number of shares purchased and the number of shares sold, following Piotroski and Roulstone (2005). I set $Sell/(Sell + Buy)_t$ to missing in years in which the manager does not trade shares. Next, I compute $NetSell_t^{Firm}$, $Sell_t^{Firm}$, and Buy_t^{Firm} by aggregating the $NetSell_t$, $Sell_t$, and Buy_t of all managers in a firm, respectively, and I take the average value of $Sell/(Sell + Buy)_t$ across all managers within a firm to obtain $Sell/(Sell + Buy)_t^{Firm}$. Lastly, I define $OptExer_t$ as the number of shares acquired through an option

¹⁰ Sias and Whidbee (2010) define *NetInsiderDemand* as insider purchases minus sales over the number of shares outstanding, which measures net purchases of shares. To be consistent with the issuance tests, I compute net sales instead of net purchases.

Table I
Summary Statistics

This table reports summary statistics. The sample consists of manufacturing firms (SICs 2000 to 3999) over the period from 2003 through 2015. I exclude firms that never report POs during the sample period. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution.

	Count	Mean	<i>SD</i>	p25	p50	p75
PO_t	10402	0.090	0.132	0.011	0.047	0.109
PO_{next_t}	10402	0.073	0.107	0.008	0.038	0.092
PO_{rest_t}	10402	0.017	0.060	0.000	0.000	0.006
ΔPO_t	10402	-0.001	0.088	-0.006	0.000	0.006
ΔPO_t^{t+1}	10402	-0.000	0.078	-0.009	0.000	0.009
ΔPO_t^{rest}	10402	-0.001	0.045	-0.002	0.000	0.001
$NetEquityIssue_t$	10402	0.017	0.131	-0.010	0.000	0.007
$NetDebtIssue_t$	10402	0.011	0.099	-0.010	0.000	0.006
$EquityIssueD_t$	10402	0.135	0.342	0.000	0.000	0.000
$RepurchaseD_t$	10402	0.183	0.387	0.000	0.000	0.000
$DebtIssueD_t$	10402	0.203	0.402	0.000	0.000	0.000
$DebtReductionD_t$	10402	0.191	0.393	0.000	0.000	0.000
$Tangibility_t$	10402	0.196	0.158	0.078	0.153	0.272
$Investment_t$	10402	0.339	0.539	0.120	0.199	0.352
$Cash_t$	10402	0.244	0.231	0.062	0.169	0.359
$Leverage_t$	10402	0.283	0.439	0.004	0.214	0.432
$Salesgrowth_t$	10402	0.202	0.806	-0.024	0.078	0.220
$Size_t$	10402	6.411	1.928	4.947	6.316	7.749
$Tobins'Q_t$	10402	2.146	1.645	1.236	1.655	2.448
$Cashflow_t$	10402	0.067	0.263	0.041	0.116	0.180
$BHAR(-1,1)$	14911	-0.000	0.037	-0.018	-0.000	0.017
$BHAR_t$	14911	0.011	0.355	-0.214	-0.014	0.190
$AnalystCoverage_t$	10402	3.910	5.756	0.000	1.000	6.182
Age_t	10402	24.815	16.856	12.000	19.000	34.000
$R\&D_t$	10402	0.191	0.357	0.016	0.064	0.172
$PublicDebtIssue_t$	5526	0.014	0.153	-0.005	0.000	0.000
$PrivateDebtIssue_t$	5526	-0.003	0.134	-0.025	0.000	0.018
DEF_t	10402	0.032	0.216	-0.056	0.000	0.051
$NetSell_t$	59006	0.234	0.991	0.000	0.000	0.000
$Sell_t$	59006	0.251	0.989	0.000	0.000	0.000
Buy_t	59006	0.017	0.124	0.000	0.000	0.000
$Sell/(Sell + Buy)_t$	12228	0.838	0.366	1.000	1.000	1.000
$NetSell_t^{Firm}$	9687	1.392	3.035	0.000	0.010	1.498
$Sell_t^{Firm}$	9687	1.498	3.002	0.000	0.029	1.532
Buy_t^{Firm}	9687	0.101	0.342	0.000	0.000	0.000
$Sell/(Sell + Buy)_t^{Firm}$	6177	0.773	0.396	0.667	1.000	1.000
$OptExer_{CEO}^t$	5703	0.133	0.235	0.000	0.000	0.175
$OptExer_t^{All}$	27349	0.174	0.289	0.000	0.000	0.235
$OptExer_t^{Firm}$	5703	0.167	0.219	0.000	0.083	0.243
$SEOminusREPO_t^{SDC}$	3875	0.029	0.077	0.000	0.000	0.000
$SEOminusREPO_t^{Cpst}$	3875	0.020	0.082	-0.003	0.000	0.000
$SEOD$	3875	0.157	0.364	0.000	0.000	0.000

exercise (Execucomp item *opt_exer_num*) divided by the sum of the number of shares acquired through an option exercise and unexercised exercisable options (*opt_unex_exer_num*), and $OptExer_t^{Firm}$ as the average value of $OptExer_t$ across all managers within a firm.

II. The Validity of ΔPO_t as a Proxy for Information Asymmetry–Driven Mispricing

In this section, I examine the two conditions that need to be satisfied to validate the use of ΔPO_t as a proxy for information asymmetry–driven mispricing.

A. Do Predisclosed POs Predict Future Stock Prices?

To test the first condition that disclosed POs predict future security prices I examine the relationship between ΔPO_t and future abnormal stock returns. To qualify as a proxy for a manager's superior information, ΔPO_t should be positively associated with abnormal stock returns following the disclosure of POs. To test this relationship, I estimate the following regression model, where *FirmFE*, *YearFE*, and *Ind-YearFE* indicate firm, year, and industry-year fixed effects, respectively, and standard errors are clustered at the firm level:

$$BHAR_{i,t+1} = \alpha_0 + \alpha_1 \Delta PO_{i,t} + FirmFE + YearFE(Ind-YearFE) + \varepsilon_{i,t}. \quad (3)$$

The results of the regression analysis are presented in Table II. The dependent variable is $BHAR_{t+1}$ in all columns. The main variables of interest are ΔPO_t in columns (1) to (4) and ΔPO_t^{t+1} and ΔPO_t^{rest} in columns (5) to (8). Because a number of studies find evidence of postearnings announcement drift, that is, a tendency for a stock's cumulative abnormal returns to drift in the direction of an earnings surprise for several weeks following an earnings announcement, I control for earnings surprises in columns (3), (4), (7), and (8). Following Kinney, Burgstahler, and Martin (2002), I define earnings surprises (ES_t) as actual earnings minus the last individual analyst forecast, scaled by price. In odd-numbered columns I control for firm and year fixed effects. To control for time-varying industry effects, I control for industry-year fixed effects in even-numbered columns.

Consistent with the view that disclosed POs predict future stock prices, the coefficient on ΔPO_t is positive and statistically significant, as reported in columns (1) to (4). In addition, including ES_t as a control variable does not reduce the statistical or economic significance of the coefficient on ΔPO_t . This result suggests that ΔPO_t contains information over and above earnings surprises. For example, the coefficient estimate shown in column (4) indicates that a one-standard-deviation increase in ΔPO_t is associated with an increase in $BHAR_{t+1}$ of approximately 1.8 percentage points, which accounts for 5.1% of the standard deviation of $BHAR_{t+1}$. The estimate also implies that an increase

Table II
 ΔPO_t and Future Stock Returns

This table provides evidence that POs predict stock prices in the following year. The dependent variable is $BHAR_{t+1}$, which is defined as size and BM-adjusted buy-and-hold returns during $t+1$. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	$BHAR_{t+1}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	0.208** (0.076)	0.176** (0.076)	0.227*** (0.078)	0.204** (0.076)				
ΔPO_t^{t+1}					0.291** (0.110)	0.320*** (0.076)	0.309** (0.114)	0.349*** (0.077)
ΔPO_t^{rest}					0.179* (0.101)	0.125 (0.104)	0.198* (0.103)	0.154 (0.107)
ES_t			0.075 (0.062)	0.066 (0.056)			0.076 (0.062)	0.067 (0.056)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Ind-year FE	No	Yes	No	Yes	No	Yes	No	Yes
N	14,911	14,911	8,066	8,066	14,911	14,911	8,066	8,066
R^2	0.211	0.247	0.212	0.248	0.211	0.248	0.212	0.249

in ΔPO_t from the 25th percentile to the 75th percentile increases $BHAR_{t+1}$ by 0.3 percentage points.

When I separate ΔPO_t^{t+1} and ΔPO_t^{rest} in columns (5) to (8), the coefficient on ΔPO_t^{t+1} is statistically and economically significant in all columns, whereas the coefficient on ΔPO_t^{rest} is statistically significant at the 10% level in columns (5) and (7). This could reflect the fact that a significant portion of PO_t is concentrated in PO_t^{t+1} , with many firms not reporting PO_t^{rest} . The results shown in column (8) indicate that a one-standard-deviation increase in ΔPO_t^{t+1} is associated with an increase in $BHAR_{t+1}$ of approximately 2.7 percentage points, which accounts for approximately 7.7% of the standard deviation of $BHAR_{t+1}$. The estimate also implies that an increase in ΔPO_t^{t+1} from the 25th percentile to the 75th percentile increases $BHAR_{t+1}$ by 0.6 percentage points. The results in Table II indicate that ΔPO_t and ΔPO_t^{t+1} are positively associated with stock returns during $t+1$. As such, the first condition is satisfied.

B. Are POs Private Information Prior to Disclosure?

To examine the second condition that predisclosed POs constitute private information I analyze stock price responses during a short time period around the disclosure of POs. If predisclosed POs are private information, stock prices should respond immediately to the disclosure of PO information. In contrast, if

Table III
 ΔPO_t and Short-Term Stock Returns

This table reports results on short-term stock price responses to the disclosure of POs. The dependent variable is $BHAR(-1, 1)$, which is defined as size and BM-adjusted buy-and-hold returns during a three-day window starting from one day before the 10-K filing date. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	<i>BHAR</i> (−1, 1)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	0.033** (0.013)	0.033** (0.013)	0.032** (0.013)	0.032** (0.013)				
ΔPO_t^{t+1}					0.041** (0.018)	0.043** (0.018)	0.041** (0.018)	0.042** (0.019)
ΔPO_t^{rest}					−0.007 (0.024)	−0.011 (0.025)	−0.008 (0.025)	−0.012 (0.025)
ES_t			0.130*** (0.037)	0.129*** (0.039)			0.130*** (0.037)	0.128*** (0.039)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Ind-year FE	No	Yes	No	Yes	No	Yes	No	Yes
<i>N</i>	14,911	14,911	8,066	8,066	14,911	14,911	8,066	8,066
<i>R</i> ²	0.211	0.247	0.212	0.248	0.211	0.248	0.212	0.249

POs are public information before disclosure, stock prices should not respond significantly following disclosure.

The results of the analysis are presented in Table III. The dependent variable is $BHAR(-1,1)$ in all columns. The main variables of interest are ΔPO_t in columns (1) to (4) and ΔPO_t^{t+1} and ΔPO_t^{rest} in columns (5) to (8). I control for earnings surprises to address postearnings announcement drift in columns (3), (4), (7), and (8). I control for firm fixed effects in all columns, year fixed effects in odd-numbered columns, and industry-year fixed effects in even-numbered columns.

Consistent with the view that POs become public information following the disclosure of 10-Ks, the coefficients on ΔPO_t are all positive and statistically significant. Although the coefficient on ES_t is positive and statistically significant, as shown in columns (3) and (4), including ES_t as a control variable does not change the statistical or economic significance of the coefficient on ΔPO_t . For example, the coefficient estimate shown in column (4) indicates that a one-standard-deviation increase in ΔPO_t is associated with an increase in $BHAR(-1,1)$ of approximately 28 bps, which accounts for 7.6% of the standard deviation of $BHAR(-1,1)$. The estimate also implies that an increase in ΔPO_t from the 25th percentile to the 75th percentile increases $BHAR(-1,1)$ by 4 bps.

I separate ΔPO_t into ΔPO_t^{t+1} and ΔPO_t^{rest} in columns (5) to (8). In all columns, only ΔPO_t^{t+1} has a statistically and economically significant

coefficient. The results shown in column (8) indicate that a one-standard-deviation increase in ΔPO_t^{t+1} is associated with an increase in $BHAR(-1,1)$ of approximately 33 bps, which accounts for 8.9% of the standard deviation of $BHAR(-1,1)$. The estimate also implies that an increase in ΔPO_t^{t+1} from the 25th percentile to the 75th percentile increases $BHAR(-1,1)$ by 8 bps.¹¹ In contrast, none of the coefficients on ΔPO_t^{rest} is positive or significant.

In sum, the results in this table support the view that prediscovered POs are private information. In addition, they help rule out an alternative explanation whereby the results presented in Table II are driven by compensation for risk. The delayed price reactions suggest, however, that investors do not fully account for adverse selection on the part of managers when security issuance decisions are made. The overall results shown in Tables II and III suggest that prediscovered changes in POs are private and superior information that managers have about firm value.

III. The Effects of Changes in POs on Security Issuance

In this section, I investigate the effects of information asymmetry-driven mispricing on security issuance using ΔPO_t as a proxy for information asymmetry-driven mispricing.¹²

A. Security Issuance

To provide evidence on the relationship between information asymmetry-driven mispricing and contemporaneous net equity and net debt issuance, I estimate the regression model presented in equation (4). I control for firm characteristics known to be associated with security issuance. To account for recent changes in fundamentals, I also control for contemporaneous and one-year-lagged abnormal stock returns in all regressions. Firm fixed effects are included in all regressions and standard errors are clustered at the firm level. I additionally control for year or industry-year fixed effects. The model is specified as follows:

$$\begin{aligned} NetEquityIssue_{i,t}(NetDebtIssue_{i,t}) = & \beta_0 + \beta_1 \Delta PO_{i,t} + \beta_2 Controls_{i,t-1} \\ & + FirmFE + \varepsilon_{i,t}. \end{aligned} \quad (4)$$

¹¹ The economic magnitude of the association between ΔPO_t and $BHAR_{t+1}$ is larger than that between ΔPO_t and $BHAR(-1,1)$, suggesting an initial underreaction of stock prices to ΔPO_t . This result is consistent with a number of findings on underreaction of stock prices to new information. For example, previous studies document underreaction of stock prices to information about stock splits (Ikenberry and Ramnath (2002)), dividend initiations (Michaely, Thaler, and Womack (1995)), spinoffs (Desai and Jain (1999)), analyst forecast revisions (Chan, Jegadeesh, and Lakonishok (1996)), and changes in reporting practices (Cohen, Malloy, and Nguyen (2020)).

¹² I use alternative measures of ΔPO_t in Appendix B.

The regression results are reported in Panel A of Table IV. The dependent variable is *NetEquityIssue_t* in columns (1) to (4). The variables of interest are ΔPO_t in columns (1) and (2) and ΔPO_t^{t+1} and ΔPO_t^{rest} in columns (3) and (4). The coefficient on ΔPO_t is negative and statistically significant, as reported in column (1). Because increases (decreases) in POs indicate undervaluation (overvaluation), this result is consistent with undervaluation (overvaluation) decreasing (increasing) contemporaneous net equity issuance. The coefficient estimate remains similar, as reported in column (2), after controlling for industry-year fixed effects. In particular, the coefficient of interest in column (2), -0.129 , suggests that a one-standard-deviation decrease in ΔPO_t is associated with an increase in contemporaneous net equity issuance of 1.1% of total securities outstanding, which accounts for approximately 8.7% of the sample standard deviation of *NetEquityIssue*. Another economic implication of this result is that a decrease in ΔPO_t from the 75th percentile to the 25th percentile is expected to increase equity issuance by 0.2% of total securities outstanding. The results shown in columns (3) and (4) suggest that this effect is due to ΔPO_t^{t+1} . The coefficient estimate in column (4), -0.230 , suggests that a one-standard-deviation decrease in ΔPO_t^{t+1} is associated with an increase in contemporaneous net equity issuance of 1.8% of the value of total securities outstanding, which accounts for 13.8% of the sample standard deviation of *NetEquityIssue*. The estimate also implies that a decrease in ΔPO_t^{t+1} from the 75th percentile to the 25th percentile increases equity issuance by 0.4% of total securities outstanding. In contrast, the coefficient on ΔPO_t^{rest} is insignificant. The coefficient on *BHAR_t* is positive and statistically significant at the 1% level in all columns. The coefficient on *BHAR_{t-1}* is positive and generally statistically significant in all columns. These results suggest that firms tend to issue equity following stock price run-ups.

In columns (5) to (8) the dependent variable is *NetDebtIssue_t*. The coefficient on ΔPO_t is negative and statistically significant, as reported in columns (5) and (6). However, the economic magnitude of the coefficients is significantly smaller than those shown in columns (1) and (2). The coefficient of interest in column (6), -0.026 , suggests that a one-standard-deviation decrease in ΔPO_t is associated with an increase in contemporaneous net debt issuance of 0.23% of total security outstanding, which accounts for approximately 2.3% of the sample standard deviation of *NetDebtIssue*. The results shown in columns (7) and (8) suggest that these effects are due to ΔPO_t^{t+1} . The coefficient estimate in column (8), -0.045 , suggests that a one-standard-deviation decrease in ΔPO_t^{t+1} is associated with an increase in contemporaneous net debt issuance of 0.35% of the value of total securities outstanding, which accounts for approximately 3.5% of the sample standard deviation of net debt issuance. Another economic implication of this result is that a decrease in ΔPO_t^{t+1} from the 75th percentile to the 25th percentile is expected to increase debt issuance by 0.08% of total securities outstanding. As such, the effects of changes in POs are weaker on debt issuance than on equity issuance. These results are consistent with the price of debt being less sensitive to information than the price of equity. The coefficient on ΔPO_t^{rest} is insignificant in both columns. Overall, the results in

Table IV
Net Security Issuance

Panel A reports results on the association between changes in POs and contemporaneous security issuance. The dependent variable is *NetEquityIssue_{it}* in columns (1) to (4) and *NetDebtIssue_{it}* in columns (5) to (8). Panel B reports results of regressions of *EquityIssue_{it}*, *Repurchase_{it}*, *DebtIssue_{it}*, and *DebtReduction_{it}*. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Panel A. Net issuance							
	<i>NetEquityIssue_{it}</i>				<i>NetDebtIssue_{it}</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	-0.127*** (0.047)	-0.129*** (0.049)			-0.026*** (0.008)	-0.026*** (0.008)		
ΔPO_t^{+1}			-0.235*** (0.057)	-0.230*** (0.058)			-0.045*** (0.011)	-0.045*** (0.011)
ΔPO_t^{rest}			-0.003 (0.006)	-0.002 (0.006)			-0.001 (0.002)	-0.001 (0.002)
<i>Tangibility_{t-1}</i>	-0.027 (0.045)	-0.027 (0.046)	-0.036 (0.028)	-0.032 (0.028)	0.052*** (0.017)	0.053*** (0.017)	0.053*** (0.016)	0.053*** (0.016)
<i>Investment_{t-1}</i>	0.002 (0.004)	0.002 (0.004)	0.001 (0.003)	0.001 (0.003)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
<i>Cash_{t-1}</i>	-0.139*** (0.023)	-0.141*** (0.023)	-0.138*** (0.016)	-0.138*** (0.016)	0.002 (0.007)	0.003 (0.007)	0.003 (0.007)	0.004 (0.007)
<i>Leverage_{t-1}</i>	-0.009 (0.015)	-0.010 (0.015)	-0.001 (0.009)	-0.001 (0.009)	-0.020*** (0.004)	-0.020*** (0.004)	-0.020*** (0.004)	-0.020*** (0.004)
<i>Salesgrowth_{t-1}</i>	-0.001 (0.003)	-0.001 (0.003)	-0.001 (0.002)	-0.001 (0.002)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
<i>Size_{t-1}</i>	-0.060*** (0.008)	-0.061*** (0.008)	-0.073*** (0.006)	-0.075*** (0.006)	-0.016*** (0.002)	-0.016*** (0.002)	-0.016*** (0.002)	-0.016*** (0.002)
<i>Tobins' Q_{t-1}</i>	-0.005*** (0.002)	-0.004** (0.002)	-0.004*** (0.001)	-0.004*** (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
<i>Cashflow_{t-1}</i>	-0.023 (0.018)	-0.025 (0.019)	-0.014 (0.011)	-0.016 (0.011)	0.015*** (0.005)	0.014*** (0.005)	0.015*** (0.005)	0.014*** (0.005)

(Continued)

Table IV—Continued

	Panel A. Net issuance							
	Net Equity Issue _t				Net Debt Issue _t			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
BHAR _t	0.021*** (0.005)	0.022*** (0.005)	0.028*** (0.006)	0.028*** (0.006)	−0.007*** (0.001)	−0.007*** (0.001)	−0.007*** (0.001)	−0.007*** (0.001)
BHAR _{t−1}	0.004* (0.002)	0.004 (0.002)	0.003** (0.001)	0.003* (0.001)	−0.001 (0.001)	−0.001 (0.001)	−0.001 (0.001)	−0.001 (0.001)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Ind-year FE	No	Yes	No	Yes	No	Yes	No	Yes
N	10,402	10,402	10,402	10,402	10,402	10,402	10,402	10,402
R ²	0.338	0.351	0.365	0.374	0.184	0.204	0.186	0.206

(Continued)

Table IV—Continued

	Panel B. Indicator variables							
	EquityIssueD _t		RepurchaseD _t		DebtIssueD _t		DebtReductionD _t	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	-0.348*** (0.097)	-0.344*** (0.099)	0.291** (0.114)	0.304*** (0.114)	-0.091** (0.045)	-0.103** (0.044)	0.066 (0.047)	0.053 (0.048)
$Tangibility_{t-1}$	-0.141* (0.077)	-0.131* (0.078)	-0.065 (0.075)	-0.025 (0.074)	0.369*** (0.101)	0.364*** (0.102)	-0.179** (0.089)	-0.170* (0.089)
$Investment_{t-1}$	0.004 (0.009)	0.005 (0.009)	-0.008* (0.005)	-0.008 (0.005)	0.006 (0.007)	0.007 (0.008)	-0.022*** (0.007)	-0.021*** (0.007)
$Cash_{t-1}$	-0.328*** (0.045)	-0.329*** (0.046)	0.179*** (0.042)	0.186*** (0.042)	-0.098** (0.041)	-0.091** (0.042)	-0.183*** (0.042)	-0.178*** (0.043)
$Leverage_{t-1}$	0.006 (0.012)	0.006 (0.012)	-0.037*** (0.009)	-0.036*** (0.009)	-0.062*** (0.017)	-0.060*** (0.017)	0.112*** (0.022)	0.112*** (0.022)
$Salesgrowth_{t-1}$	-0.001 (0.007)	-0.001 (0.007)	-0.004** (0.002)	-0.004** (0.002)	-0.001 (0.005)	-0.001 (0.005)	0.001 (0.004)	0.002 (0.004)
$Size_{t-1}$	-0.118*** (0.013)	-0.123*** (0.013)	0.034*** (0.011)	0.037*** (0.011)	-0.040*** (0.013)	-0.041*** (0.013)	0.075*** (0.012)	0.075*** (0.013)
$Tobins' Q_{t-1}$	0.009** (0.004)	0.008** (0.004)	-0.005* (0.003)	-0.004 (0.003)	-0.002 (0.004)	-0.001 (0.004)	-0.012*** (0.003)	-0.013*** (0.003)
$Cash\ flow_{t-1}$	-0.101*** (0.031)	-0.101*** (0.031)	0.088*** (0.018)	0.084*** (0.018)	0.028 (0.032)	0.028 (0.032)	-0.070*** (0.023)	-0.065*** (0.023)
$BHAR_t$	0.060*** (0.007)	0.060*** (0.007)	-0.008* (0.004)	-0.005 (0.004)	-0.017*** (0.006)	-0.019*** (0.007)	0.053*** (0.007)	0.053*** (0.007)
$BHAR_{t-1}$	0.020*** (0.006)	0.021*** (0.006)	-0.006 (0.004)	-0.006 (0.004)	-0.004 (0.006)	-0.003 (0.006)	0.012* (0.006)	0.011* (0.006)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Ind-year FE	No	Yes	No	Yes	No	Yes	No	Yes
N	10,402	10,402	10,402	10,402	10,402	10,402	10,402	10,402
R ²	0.423	0.436	0.380	0.402	0.213	0.234	0.287	0.307

Panel A suggest that a manager's perception of mispricing strongly influences security issuance, especially equity issuance.

I next examine the association between ΔPO_t and the propensity to issue or reduce equity and debt. In particular, I run regressions in which the dependent variable is *EquityIssueD*, *RepurchaseD*, *DebtIssueD*, and *DebtReductionD*. The results are presented in Panel B. The dependent variable in columns (1) and (2) is *EquityIssueD_t*. The coefficient on ΔPO_t in column (1), -0.348 , is statistically significant at the 1% level. This coefficient estimate indicates that a one-standard-deviation decrease in ΔPO_t increases the probability of equity issuance by 3.1 percentage points, which accounts for approximately 22.7% of the sample mean of *EquityIssueD_t*. The result also suggests that a decrease in ΔPO_t from the 75th percentile to the 25th percentile is expected to increase the probability of equity issuance by 0.4 percentage points. The results remain similar when I control for industry-year fixed effects in column (2). The coefficients on *BHAR_t* and *BHAR_{t-1}* are positive and statistically significant, suggesting that recent stock performance is positively associated with equity issuance. The dependent variable in columns (3) and (4) is *RepurchaseD_t*. The coefficient on ΔPO_t is statistically significant and positive in both columns. The estimated coefficient reported in column (4), 0.304 , indicates that a one-standard-deviation increase in ΔPO_t increases the probability of equity repurchase by 2.7 percentage points, which accounts for approximately 14.6% of the sample mean of *RepurchaseD_t*. The estimate also implies that an increase in ΔPO_t^{t+1} from the 25th percentile to the 75th percentile is expected to increase the probability of equity repurchases by 0.4 percentage points.

The dependent variable in columns (5) and (6) is *DebtIssueD_t*. The coefficient on ΔPO_t is negative and statistically significant at the 5% level. The magnitudes of the coefficients, however, are approximately one-third of those shown in columns (1) and (2). In particular, the coefficient estimate in column (6) indicates that a one-standard-deviation decrease in ΔPO_t increases the probability of debt issuance by 0.9 percentage points, which accounts for approximately 4.5% of the sample mean of *DebtIssueD_t*. The result also suggests that a decrease in ΔPO_t from the 75th percentile to the 25th percentile is expected to increase the probability of debt issuance by 0.1 percentage points. Finally, the dependent variable in columns (7) and (8) is *DebtReductionD_t*. The coefficients on ΔPO_t are positive but statistically and economically insignificant, as reported in both columns. This result suggests that managers do not actively reduce outstanding debt when POs predict undervaluation of their firms. Thus, information asymmetry-driven mispricing affects debt issuance but not debt reductions, while on the equity side it affects both issues and repurchases of equity. In addition, the magnitude of the effects of information asymmetry-driven mispricing on security issues is significantly stronger for equity. These results are again consistent with the price of debt being less sensitive to information than the price of equity.

Table V
Financial Leverage

This table reports results on the association between changes in POs and leverage. The dependent variable is *Leverage*. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	<i>Leverage_t</i>			
	(1)	(2)	(3)	(4)
ΔPO_t	0.059** (0.024)	0.071*** (0.024)		
ΔPO_t^{t+1}			0.062** (0.028)	0.073** (0.030)
ΔPO_t^{rest}			0.054* (0.031)	0.066** (0.032)
<i>Tangibility</i> _{<i>t</i>−1}	−0.011 (0.113)	−0.017 (0.118)	−0.011 (0.125)	−0.017 (0.129)
<i>Investment</i> _{<i>t</i>−1}	−0.024*** (0.008)	−0.024*** (0.008)	−0.024** (0.010)	−0.024** (0.010)
<i>Cash</i> _{<i>t</i>−1}	−0.213*** (0.051)	−0.222*** (0.056)	−0.213*** (0.064)	−0.222*** (0.066)
<i>Salesgrowth</i> _{<i>t</i>−1}	0.001 (0.005)	0.001 (0.005)	0.001 (0.010)	0.001 (0.010)
<i>Size</i> _{<i>t</i>−1}	0.001 (0.018)	−0.001 (0.017)	0.001 (0.019)	−0.001 (0.020)
<i>Tobins'Q</i> _{<i>t</i>−1}	−0.013** (0.006)	−0.012* (0.007)	−0.013* (0.007)	−0.012* (0.007)
<i>Cashflow</i> _{<i>t</i>−1}	−0.062 (0.042)	−0.053 (0.040)	−0.062 (0.050)	−0.053 (0.050)
<i>BHAR</i> _{<i>t</i>}	−0.036*** (0.011)	−0.036*** (0.010)	−0.036*** (0.013)	−0.036*** (0.013)
<i>BHAR</i> _{<i>t</i>−1}	−0.031*** (0.009)	−0.032*** (0.009)	−0.031*** (0.010)	−0.032*** (0.010)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Ind-Year FE	No	Yes	No	Yes
<i>N</i>	10,402	10,402	10,402	10,402
<i>R</i> ²	0.514	0.524	0.514	0.524

B. Financial Leverage

I next investigate the consequences of information asymmetry-driven mispricing for financial leverage. The results are presented in Table V. The dependent variable is *Leverage_t* in all columns. The coefficient on ΔPO_t is significant and positive, as reported in column (1), indicating a positive association between changes in POs and contemporaneous leverage. To control for time-varying industry effects of leverage (Lemmon, Roberts, and Zender (2008)), I

control for industry-year fixed effects in column (2). The coefficient of interest remains at a similar level. In columns (3) and (4), I separate ΔPO_t into ΔPO_t^{t+1} and ΔPO_t^{rest} . The coefficients on ΔPO_t^{t+1} and ΔPO_t^{rest} are significant and positive, as reported in both columns. However, the coefficient on ΔPO_t^{t+1} is statistically and economically more significant than that on ΔPO_t^{rest} . Overall, the results in this table suggest that information asymmetry-driven undervaluation (overvaluation) has a positive (negative) effect on leverage, consistent with the results for security issuance in Table IV.

IV. The Level of Information Asymmetry and the Effects of ΔPO_t

To test the validity of the interpretation that the negative association between ΔPO_t and security issuance is due to information asymmetry-driven mispricing, I conduct additional analyses in which I exploit variation in the level of information asymmetry. I first exploit cross-sectional variation in the level of information asymmetry and examine whether the association is stronger among firms with higher levels of information asymmetry. I then exploit time-varying information asymmetry and examine whether the timing of issuance is consistent with the information asymmetry-based explanation. I also exploit variation in levels of information-sensitivity between public debt and private debt, and examine whether issuance of information-sensitive debt is more negatively associated ΔPO_t .

A. Cross-Sectional Analyses

Here, I repeat the *NetEquityIssue* and *NetDebtIssue* analyses after categorizing firms based on the level of information asymmetry. If information asymmetry-driven mispricing is behind my findings, the association between ΔPO_t and *NetEquityIssue_t* (*NetDebtIssue_t*) should be stronger among firms with high levels of information asymmetry. In contrast, if the association is driven by factors other than information-asymmetry problems, there is no clear reason for the association to be stronger among firms with high information asymmetry. I first exploit variation in the extent to which POs comprise private information. In particular, I separately analyze a small subset of sample firms that occasionally report POs on 10-Qs during the sample period. The level of information asymmetry due to prediscovered POs will be lower for these firms compared with firms that report POs only on annual reports. Next, I employ the number of analysts following, firm age, and R&D intensity to separate sample firms. Low (high) analyst following, young (old) firms, and high (low) R&D-intensity firms are categorized into high- (low-) information-asymmetry groups.

I report the results in Table VI. The dependent variable is *NetEquityIssue_t* in Panel A. In column (1) I report the results of the regression in which I use firms that regularly report POs in their 10-Q filings during the sample period. Approximately 10% of sample firms occasionally report POs on 10-Qs even in the

Table VI
Level of Information Asymmetry

This table presents results on security issuance, where I categorize firms into high/low-information-asymmetry groups. The dependent variable is $NetEquityIssue_t$ in Panel A and $NetDebtIssue_t$ in Panel B. In column (1), I separately analyze a small subset of sample firms that occasionally report POs on 10-Qs during the sample period. In columns (2) and (3) I report results of regressions using high and low analyst coverage, respectively. In columns (4) and (5) I report results of regressions for old and young firms, respectively. Columns (6) and (7) present results of regressions for firms with low and high R&D intensity, respectively. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A. <i>NetEquityIssue_{it}</i>							
	10Q	Analyst Coverage		Age		R&D	
	Yes (1)	High (2)	Low (3)	Old (4)	Young (5)	Low (6)	High (7)
ΔPO_t	-0.056 (0.046)	-0.068*** (0.022)	-0.111*** (0.036)	-0.061** (0.025)	-0.312*** (0.101)	-0.019 (0.019)	-0.172*** (0.036)
$Tangibility_{t-1}$	-0.012 (0.052)	-0.034 (0.023)	-0.012 (0.026)	-0.002 (0.034)	-0.066 (0.061)	-0.046* (0.025)	-0.013 (0.026)
$Investment_{t-1}$	0.000 (0.005)	0.004 (0.004)	0.001 (0.002)	0.014 (0.014)	0.001 (0.004)	0.006 (0.006)	0.001 (0.002)
$Cash_{t-1}$	-0.088*** (0.027)	-0.048*** (0.011)	-0.097*** (0.016)	-0.073*** (0.020)	-0.170*** (0.035)	-0.091*** (0.016)	-0.081*** (0.013)
$Leverage_{t-1}$	0.036*** (0.012)	0.017 (0.011)	-0.005 (0.004)	0.005 (0.013)	-0.004 (0.017)	-0.010 (0.006)	0.007 (0.004)
$Salesgrowth_{t-1}$	-0.002 (0.005)	-0.001 (0.004)	0.000 (0.002)	0.000 (0.008)	-0.001 (0.003)	0.002 (0.003)	-0.000 (0.002)
$Size_{t-1}$	-0.027*** (0.009)	-0.023*** (0.004)	-0.045*** (0.005)	-0.024*** (0.005)	-0.082*** (0.013)	-0.018*** (0.004)	-0.046*** (0.004)
$Tobins' Q_{t-1}$	0.001 (0.002)	-0.002 (0.001)	-0.002* (0.001)	-0.003 (0.003)	-0.005** (0.002)	0.002 (0.001)	-0.003*** (0.001)
$Cashflow_{t-1}$	0.004 (0.022)	-0.022 (0.014)	-0.004 (0.009)	-0.034 (0.024)	-0.019 (0.023)	-0.048*** (0.016)	0.001 (0.009)

(Continued)

Table VI—Continued

	Panel A. <i>NetEquityIssue_{it}</i>						
	Analyst Coverage			Age		R&D	
	10Q Yes (1)	High (2)	Low (3)	Old (4)	Young (5)	Low (6)	High (7)
<i>BHAR_t</i>	0.013** (0.006)	0.014*** (0.003)	0.011*** (0.003)	0.010** (0.004)	0.028*** (0.008)	0.006*** (0.002)	0.018*** (0.003)
<i>BHAR_{t-1}</i>	0.011* (0.007)	0.005** (0.002)	0.003* (0.002)	0.006 (0.005)	0.004 (0.002)	−0.001 (0.001)	0.008*** (0.002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,015	5,277	5,125	5,248	5,154	5,217	5,185
<i>R</i> ²	0.479	0.413	0.492	0.308	0.355	0.365	0.524

(Continued)

Table VI—Continued

	Panel B. Net Debt Issue _t						
	10Q	Analyst Coverage		Age		R&D	
	Yes (1)	High (2)	Low (3)	Old (4)	Young (5)	Low (6)	High (7)
ΔPO_t	-0.042 (0.026)	-0.023** (0.011)	-0.030** (0.012)	-0.031** (0.012)	-0.057* (0.030)	-0.016 (0.011)	-0.039*** (0.012)
$Tangibility_{t-1}$	0.001 (0.058)	0.015 (0.024)	0.067*** (0.023)	-0.003 (0.043)	0.107** (0.043)	0.044* (0.025)	0.067*** (0.022)
$Investment_{t-1}$	0.001 (0.004)	-0.000 (0.003)	0.001 (0.001)	0.007 (0.004)	-0.001 (0.002)	-0.001 (0.005)	0.000 (0.001)
$Cash_{t-1}$	-0.011 (0.024)	-0.006 (0.011)	0.004 (0.009)	0.020 (0.018)	-0.020 (0.015)	0.024 (0.016)	-0.001 (0.008)
$Leverage_{t-1}$	-0.054*** (0.015)	-0.049*** (0.007)	-0.015*** (0.004)	-0.038*** (0.014)	-0.033*** (0.009)	-0.025*** (0.007)	-0.021*** (0.005)
$Salesgrowth_{t-1}$	-0.000 (0.004)	-0.003 (0.003)	0.001 (0.001)	0.002 (0.005)	0.001 (0.002)	0.001 (0.004)	-0.000 (0.001)
$Size_{t-1}$	-0.029*** (0.011)	-0.022*** (0.003)	-0.013*** (0.003)	-0.033*** (0.006)	-0.025*** (0.006)	-0.032*** (0.004)	-0.007*** (0.002)
$Tobins'Q_{t-1}$	0.001 (0.002)	-0.001 (0.001)	0.000 (0.001)	-0.002 (0.003)	-0.002 (0.001)	0.000 (0.001)	-0.000 (0.001)
$Cashflow_{t-1}$	-0.012 (0.015)	0.036*** (0.011)	0.009* (0.005)	0.075*** (0.020)	0.013 (0.009)	0.055*** (0.016)	0.004 (0.005)
$BHAR_t$	-0.011 (0.009)	-0.006*** (0.002)	-0.007*** (0.001)	-0.011*** (0.003)	-0.009*** (0.002)	-0.008*** (0.002)	-0.006*** (0.001)
$BHAR_{t-1}$	-0.010* (0.006)	-0.001 (0.002)	-0.001 (0.001)	-0.006** (0.003)	0.002 (0.003)	-0.002 (0.002)	-0.001 (0.001)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,015	5,277	5,125	5,248	5,154	5,217	5,185
R ²	0.274	0.174	0.209	0.152	0.223	0.194	0.219

absence of material changes.¹³ The coefficient on ΔPO_t is negative but statistically insignificant. In addition, the magnitude of the coefficient is less than half the estimated coefficient using all sample firms. These results indicate that the negative association between ΔPO_t and $NetEquityIssue_t$ documented in Table IV is weak, if it exists at all, among these firms, consistent with the view that the negative association between ΔPO_t and $NetEquityIssue_t$ is stronger among firms with high information asymmetry.

In columns (2) and (3) I report results of regressions using firms with high and low analyst following, respectively. I categorize a firm into the high-analyst-following group if its time-series average of analyst coverage at the firm level is larger than one and into the small-analyst-coverage group otherwise.¹⁴ Although it is negative in both columns, the magnitude of the coefficient on ΔPO_t in column (4) is 63% larger than that in column (3). This result is consistent with the view that the effects of information asymmetry-driven mispricing on equity issuance are stronger among those firms that are associated with more severe information asymmetry problems.

In columns (4) and (5) I report results of the $NetEquityIssue$ regression for old and young firms, respectively. To estimate a firm's age, I use the first year it appears in the Compustat database. The coefficient on ΔPO_t is negative, as reported in both columns. However, the absolute value of the reported coefficient is significantly larger and statistically more significant in column (5). This suggests that the negative association between ΔPO_t and $NetEquityIssue_t$ is more pronounced among young firms. In columns (6) and (7) I report results of the $NetEquityIssue$ regression for firms with low and high R&D intensity, respectively, where R&D intensity is the time-series average of R&D expenditures over sales at the firm level. The coefficient on ΔPO_t is negative, as reported in both columns, but is economically and statistically significant only in column (7). This result suggests that the effects of ΔPO_t are more pronounced among high-R&D-intensity firms. Overall, the results presented in Panel A are consistent with the view that the negative association between ΔPO_t and $NetEquityIssue_t$ is stronger among firms with more severe information-asymmetry problems, and thus support the interpretation that this association is due to information asymmetry.

The dependent variable is $NetDebtIssue_t$ in Panel B. In column (1) I report results of the regression for firms that regularly report POs in their 10-Q filings during the sample period. The coefficient on ΔPO_t is statistically insignificant, although the magnitude of the coefficient is larger than that shown in Table IV, where I use all firms in the regression. In columns (2) and (3) I report results of regressions using high and low analyst coverage, respectively. Although negative in both columns, the magnitude of the coefficient on ΔPO_t is larger in

¹³ Some firms regularly mention on their 10-Qs that there were no material changes in POs without reporting POs. I do not include these firms in this analysis. A firm is included in the analysis if it reports POs on 10-Qs for over half of the quarters during the sample period. Almost all firms maintain the practice of reporting POs on 10-Qs once they first report POs on 10-Qs.

¹⁴ If a firm is not covered in I/B/E/S in a given year, I set the number of analysts to zero.

column (3), consistent with the effects of information asymmetry-driven mispricing on debt issuance being stronger among firms associated with more severe information asymmetry problems. In columns (4) and (5) I report results of the *NetDebtIssue* regression for old and young firms, respectively. The magnitude of the coefficient on ΔPO_t is larger in column (5) but the statistical significance is greater in column (4). Columns (6) and (7) present results for low- and high-R&D-intensity firms, respectively. The coefficient on ΔPO_t is negative, as reported in both columns, but is economically and statistically significant only in column (7), suggesting that the effects of ΔPO_t are more pronounced among high-R&D-intensity firms. Overall, the results in Panel B are generally consistent with the negative association between ΔPO_t and *NetDebtIssue*_{*t*} being stronger among firms with more severe information asymmetry problems. Nevertheless, the level of information asymmetry appears to play a more important role for equity issuance than debt issuance, which is again consistent with the view that the value of debt is less sensitive to information than the value of equity.

B. Time-Varying Information Asymmetry

One would expect the level of information asymmetry regarding POs between managers and outside investors to increase as time passes following the most recent disclosure. If this were the case, the association between ΔPO_t and net security issuance should be stronger during the last fiscal quarter of *t* (Q4_{*t*}). To test this prediction, I examine the association between ΔPO_t and net equity issuance (net debt issuance) during each fiscal quarter of year *t*. The results of the *NetEquityIssue* analyses are presented in Panel A of Table VII. The dependent variable is net equity issuance during Q1_{*t*}, Q2_{*t*}, Q3_{*t*}, and Q4_{*t*} in columns (1), (2), (3), and (4), respectively. The coefficient on ΔPO_t is negative and statistically significant in all four columns. However, the magnitude of the coefficient on ΔPO_t reported in column (4) is larger than those reported in columns (1) to (3). In addition, the coefficient on ΔPO_t reported in column (3) is larger than those reported in columns (1) and (2). As expected, these results are consistent with the negative association between ΔPO_t and net equity issuance being stronger as information asymmetry grows over time following the most recent 10-K filings.

I next examine whether ΔPO affects equity issuance after the information asymmetry is resolved. In particular, I analyze the association between ΔPO_{t-1} and *NetEquityIssue* during Q1_{*t*}. Because ΔPO_{t-1} is no longer private information in Q1_{*t*}, there should be no negative association between ΔPO_{t-1} and *NetEquityIssue* during Q1_{*t*} if the results in Table IV are due to information asymmetry. In contrast, the firm's fundamentals, including demand for external financing, likely remain similar from Q4_{*t-1*} through Q1_{*t*}. The results are presented in columns (5) and (6). The dependent variable is net equity issuance during Q1_{*t*}, as reported in both columns. The variable of interest is ΔPO_{t-1} in column (5). As can be seen, the coefficient on ΔPO_{t-1} is neither economically nor statistically significant. In column (6), where I add both ΔPO_{t-1} and ΔPO_t

Table VII
Quarterly Security Issuance

This table reports results on quarterly security issuance. The dependent variable is *NetEquityIssue_t* in Panel A and *NetDebtIssue_t* in Panel B. Columns (1) to (4) provide results on the association between ΔPO_t and quarterly net security issuance. Columns (5) and (6) provide results for regressions of net security issuance during $Q1_t$ on ΔPO_{t-1} . All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Panel A. <i>NetEquityIssue_t</i>					
	Q1 _t (1)	Q2 _t (2)	Q3 _t (3)	Q4 _t (4)	Q1 _t (5)	Q1 _t (6)
ΔPO_t	−0.027*** (0.009)	−0.025* (0.015)	−0.038*** (0.014)	−0.053** (0.025)		−0.021** (0.009)
ΔPO_{t-1}					−0.009 (0.011)	0.001 (0.011)
<i>Tangibility</i> _{t−1}	0.003 (0.009)	−0.006 (0.013)	−0.015 (0.014)	−0.030 (0.018)	−0.004 (0.009)	−0.001 (0.010)
<i>Investment</i> _{t−1}	−0.002* (0.001)	0.003 (0.002)	0.001 (0.001)	0.000 (0.003)	−0.000 (0.001)	−0.001 (0.001)
<i>Cash</i> _{t−1}	−0.033*** (0.006)	−0.038*** (0.008)	−0.029*** (0.006)	−0.023** (0.011)	−0.036*** (0.006)	−0.036*** (0.007)
<i>Leverage</i> _{t−1}	−0.002 (0.002)	0.003 (0.004)	−0.002 (0.003)	0.003 (0.007)	−0.001 (0.002)	−0.002 (0.003)
<i>Salesgrowth</i> _{t−1}	−0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	−0.001 (0.002)	−0.000 (0.001)	0.000 (0.001)
<i>Size</i> _{t−1}	−0.010*** (0.002)	−0.016*** (0.002)	−0.012*** (0.002)	−0.017*** (0.004)	−0.009*** (0.002)	−0.009*** (0.002)
<i>Tobins' Q</i> _{t−1}	0.001*** (0.001)	−0.002*** (0.001)	−0.000 (0.001)	−0.003*** (0.001)	0.001** (0.001)	0.002*** (0.001)
<i>Cashflow</i> _{t−1}	−0.005 (0.003)	−0.001 (0.006)	−0.009 (0.006)	−0.007 (0.014)	−0.009** (0.004)	−0.010** (0.004)
<i>BHAR</i> _t	−0.001 (0.001)	0.006*** (0.002)	0.004*** (0.001)	0.013*** (0.004)	−0.001 (0.001)	−0.001 (0.001)
<i>BHAR</i> _{t−1}	0.000 (0.001)	0.001 (0.001)	0.002** (0.001)	0.002 (0.001)	−0.000 (0.001)	−0.000 (0.001)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	10,402	10,402	10,402	10,402	9,876	9,044
<i>R</i> ²	0.272	0.260	0.267	0.207	0.269	0.291
	Panel B. <i>NetDebtIssue_t</i>					
	Q1 _t (1)	Q2 _t (2)	Q3 _t (3)	Q4 _t (4)	Q1 _t (5)	Q1 _t (6)
ΔPO_t	−0.004** (0.002)	−0.001 (0.002)	−0.001 (0.002)	−0.015** (0.008)		−0.004* (0.002)
ΔPO_{t-1}					0.005 (0.005)	0.005 (0.005)

(Continued)

Table VII—Continued

	Panel B. <i>NetDebtIssue_t</i>					
	Q1 _t (1)	Q2 _t (2)	Q3 _t (3)	Q4 _t (4)	Q1 _t (5)	Q1 _t (6)
<i>Tangibility_{t-1}</i>	0.016*** (0.004)	0.011*** (0.004)	0.010** (0.004)	-0.002 (0.006)	0.015*** (0.004)	0.012*** (0.003)
<i>Investment_{t-1}</i>	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.001 (0.000)	0.001* (0.000)	0.001*** (0.000)
<i>Cash_{t-1}</i>	-0.004** (0.002)	0.004** (0.002)	-0.000 (0.002)	0.000 (0.002)	-0.003** (0.002)	-0.004*** (0.001)
<i>Leverage_{t-1}</i>	-0.002** (0.001)	-0.003*** (0.001)	-0.001 (0.001)	-0.004*** (0.001)	-0.002*** (0.001)	-0.001** (0.000)
<i>Salesgrowth_{t-1}</i>	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
<i>Size_{t-1}</i>	-0.002*** (0.000)	-0.002*** (0.001)	-0.001* (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.001*** (0.000)
<i>Tobins'Q_{t-1}</i>	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)
<i>Cashflow_{t-1}</i>	0.000 (0.001)	0.002 (0.001)	0.002* (0.001)	0.002 (0.001)	0.000 (0.001)	0.000 (0.001)
<i>BHAR_t</i>	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001* (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
<i>BHAR_{t-1}</i>	-0.001*** (0.000)	-0.000 (0.000)	0.000 (0.000)	0.001 (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	10,402	10,402	10,402	10,402	9,876	9,044
<i>R</i> ²	0.240	0.199	0.185	0.198	0.242	0.241

to the regression, only the coefficient on ΔPO_t is negative and statistically significant. This result suggests that the negative association between ΔPO and equity issuance disappears as soon as ΔPO becomes public information, in line with the information-asymmetry explanation.

In Panel B I repeat the analyses above for *NetDebtIssue*. The dependent variable is net debt issuance during Q1_t, Q2_t, Q3_t, and Q4_t in columns (1), (2), (3), and (4), respectively. The coefficient on ΔPO_t is negative, as reported in all four columns, but the magnitude and statistical significance of the coefficient on ΔPO_t is largest in column (4). The magnitude of the coefficient is approximately 2.5 times the sum of the coefficients of interest reported in columns (1) to (3), but the magnitude and statistical significance of the coefficients do not increase monotonically in columns (1) to (3). For example, the coefficient of interest is statistically significant in column (1) but not in columns (2) and (3), while the magnitude of the coefficient reported in column (1) is larger than those reported in columns (2) and (3).

In columns (5) and (6), I report results of analyses in which I examine whether ΔPO affects debt issuance when it is no longer private information. The dependent variable is net debt issuance during Q1_t in both columns. The

coefficient on ΔPO_{t-1} reported in column (5) is positive but neither economically nor statistically significant. In column (6), the coefficient on ΔPO_{t-1} remains similar when ΔPO_t is added to the regression and the coefficient on ΔPO_t is negative and statistically significant at the 10% level. These results suggest that the negative association between ΔPO and debt issuance disappears when ΔPO becomes public information, again in line with the information asymmetry interpretation. Overall, the results in this table suggest that the negative association between ΔPO and security issuance is stronger when information asymmetry is more severe, especially for equity issuance.

C. Public Debt versus Private Debt

If information asymmetry drives the negative association between ΔPO_t and net debt issuance, the association should be stronger for public debt than for private debt. This is because public debt is more susceptible to information asymmetry problems than private debt, reflecting significant institutional differences between the two markets. For example, banks generally request information, both public and private, that they deem necessary to evaluate a borrower's financial health and then use that information to determine the terms of a contract. In addition, a borrower is typically required to provide periodic reports to a bank, which further reduces the costs associated with information asymmetry. Banks are also known to be better able to process information about and monitor borrowers than bondholders, who typically do not have access to information about or the ability to monitor borrowers. In addition, because public debt is typically held by dispersed arm's-length lenders, a free-rider problem arises in monitoring.

To test whether the negative association with ΔPO_t is stronger for public debt than for private debt, I take advantage of the Capital IQ database and separately examine public debt issues and private debt issues. In particular, I define *PublicDebtIssue* as bonds and notes minus lagged bonds and notes, scaled by the lagged value of total securities, in which bonds and notes are computed as the sum of senior bonds and notes and subordinated bonds and notes. I define *PrivateDebtIssue* as nonpublic debt minus lagged nonpublic debt, scaled by the lagged value of total securities.

Table VIII reports the results. I restrict the sample to firm-year observations covered by the Capital IQ database. I first check whether the results of regressions of *NetEquityIssue* and *NetDebtIssue* in Table IV hold in this sample. The dependent variables are *NetEquityIssue* and *NetDebtIssue* in columns (1) and (2), respectively. In both columns, the statistical and economic significance of the coefficient on ΔPO_t remain similar to those shown in Table IV. The dependent variable is *PublicDebtIssue* in column (3). The coefficient on ΔPO_t is negative and statistically significant. Moreover, at -0.069 , the coefficient is more than twice that reported in column (2). The dependent variable is *PrivateDebtIssue* in column (4). The coefficient on ΔPO_t is neither statistically nor economically significant. Moreover, the sign on the coefficient is not negative. These results suggest that the negative association between ΔPO_t

Table VIII
Public Debt versus Private Debt

This table reports results on the association between ΔPO_t and public (private) debt issuance. In these analyses, I use firm-years covered by the Capital IQ database. The dependent variables are *NetEquityIssue_t*, *NetDebtIssue_t*, *PublicDebtIssue_t*, and *PrivateDebtIssue_t* in columns (1), (2), (3) and (4), respectively. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	<i>NetEquity Issue_t</i> (1)	<i>NetDebt Issue_t</i> (2)	<i>PublicDebt Issue_t</i> (3)	<i>PrivateDebt Issue_t</i> (4)
ΔPO_t	-0.137*** (0.051)	-0.028*** (0.011)	-0.069** (0.030)	0.002 (0.019)
<i>Tangibility_{t-1}</i>	-0.058 (0.042)	0.037* (0.022)	0.191*** (0.061)	-0.031 (0.038)
<i>Investment_{t-1}</i>	0.000 (0.006)	0.003 (0.003)	0.010 (0.008)	-0.001 (0.007)
<i>Cash_{t-1}</i>	-0.115*** (0.027)	0.018* (0.010)	-0.063** (0.032)	0.072*** (0.020)
<i>Leverage_{t-1}</i>	-0.009 (0.013)	-0.022*** (0.006)	-0.043*** (0.008)	0.003 (0.006)
<i>Salesgrowth_{t-1}</i>	-0.000 (0.004)	-0.001 (0.001)	0.011** (0.004)	-0.006* (0.003)
<i>Size_{t-1}</i>	-0.047*** (0.007)	-0.021*** (0.003)	-0.032*** (0.008)	-0.004 (0.005)
<i>Tobins'Q_{t-1}</i>	-0.001 (0.002)	0.002* (0.001)	0.018*** (0.003)	-0.006** (0.003)
<i>Cashflow_{t-1}</i>	-0.012 (0.018)	0.029*** (0.009)	0.059** (0.024)	-0.006 (0.019)
<i>BHAR_t</i>	0.026*** (0.007)	-0.007*** (0.002)	-0.016*** (0.005)	0.005 (0.003)
<i>BHAR_{t-1}</i>	0.001 (0.003)	-0.003** (0.001)	-0.010** (0.004)	0.007** (0.003)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<i>N</i>	5,526	5,526	5,526	5,526
<i>R</i> ²	0.345	0.177	0.189	0.114

and net debt issuance is due to public debt issuance, in line with the interpretation that the negative association between ΔPO and contemporaneous net debt issuance is due to information asymmetry-driven mispricing.

V. Alternative Explanations

In this section, I address alternative explanations. First, I consider the concern that the negative association between ΔPO_t and contemporaneous security issues may be due to time-varying demand for external financing that is correlated with changes in POs. To this end, I investigate whether ΔPO_t

is associated with a contemporaneous or future financing deficit. I next address the possibility that inactive equity activities, such as scheduled employee stock grants, are driving the negative association between ΔPO_t and $NetEquityIssue_t$.

A. ΔPO_t and Demand for External Financing

Here I consider the possibility that the negative association between ΔPO_t and security issues that I document is due to time-varying fundamentals that are correlated with changes in POs. Although recent changes in fundamentals of which the market is aware will be captured in contemporaneous and one-year-lagged abnormal stock returns, ΔPO_t could capture information about demand for external financing that is over and above what the market knows. For example, a decrease (increase) in PO may indicate lower (higher) cash flows in the following year and thus a firm's need for external financing may increase (decrease). To address this possibility, I examine whether ΔPO_t is associated with a contemporaneous or future financing deficit. To this end, I follow Frank and Goyal (2003) and define DEF_t as cash dividends plus investment plus the change in working capital minus internal cash flow, scaled by lagged assets.

The results are presented in Table IX. I first examine the association between changes in POs and a contemporaneous financing deficit. The dependent variable is DEF_t and the variable of interest is ΔPO_t in columns (1) and (2). In column (1), I do not control for firm characteristics. The coefficient of interest is neither statistically nor economically significant. I control for firm characteristics in column (2) but the coefficient of interest remains qualitatively unchanged, suggesting that ΔPO_t is not significantly associated with a contemporaneous financing deficit. In columns (3) and (4), I report results on whether changes in POs are associated with future financing deficits. The dependent variable is DEF_{t+1} in both columns. Again, none of the coefficients of interest is statistically significant, indicating that changes in POs are not associated with future financing deficits. Overall, the results reported in this table mitigate the concern that ΔPO is negatively associated with demand for external financing and could drive the results presented in Table IV.

B. Alternative Measures of Equity Issuance

To mitigate the concern that inactive equity activities, such as scheduled employee stock grants, that are included in $NetEquityIssue_t$ are driving the results, I use alternative measures of net equity issuance. In particular, I define $SEOminusREPO^{SDC}$ as the difference between the dollar value of SEOs and stock repurchases, both of which come from the SDC database, scaled by lagged total securities outstanding. Because Banyl, Dyl, and Kahle (2008) find that publicly announced share repurchase programs are often not reported by SDC, and the Compustat measures provide fairly accurate

Table IX
Financing Deficit

This table reports results of regressions of financing deficit. The dependent variable is DEF_t , which is calculated as cash dividends plus investment plus the change in working capital minus internal cash flow, scaled by lagged assets. In columns (1) and (2), I examine the association between changes in POs and contemporaneous DEF . In columns (3) and (4), I examine the association between changes in POs and future DEF . All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	DEF_t		DEF_{t+1}	
	(1)	(2)	(3)	(4)
ΔPO_t	-0.019 (0.200)	-0.017 (0.202)	0.060 (0.197)	0.108 (0.197)
$Tangibility_{t-1}$		-0.106 (0.496)		-0.473 (0.408)
$Investment_{t-1}$		0.061 (0.049)		0.020 (0.050)
$Cash_{t-1}$		-0.951*** (0.298)		-0.808*** (0.264)
$Leverage_{t-1}$		-0.059 (0.071)		-0.039 (0.077)
$Salesgrowth_{t-1}$		0.016 (0.031)		0.028 (0.022)
$Size_{t-1}$		0.012 (0.064)		-0.021 (0.060)
$Tobins'Q_{t-1}$		-0.072*** (0.024)		0.022 (0.021)
$Cash\ flow_{t-1}$		0.637*** (0.154)		0.250* (0.135)
$BHAR_t$		-0.083* (0.045)		-0.010 (0.039)
$BHAR_{t-1}$		-0.009 (0.037)		-0.018 (0.034)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,402	10,402	9,030	9,030
R^2	0.181	0.184	0.188	0.190

estimates of actual total share repurchases for firms with SDC coverage, I define $SEO_{minus}REPO^{Cpst}$ alternatively as SEOs minus PRSTKC in Compustat, scaled by lagged total securities outstanding. I also define $SEOD$ as an indicator variable that is equal to one if a firm engages in an SEO in a given year and zero otherwise.

The results are presented in Table X. The dependent variable is $SEO_{minus}REPO_t^{SDC}$ in columns (1) and (2). In column (1), I report results of a regression in which I use all sample firms. The coefficient on ΔPO_t is negative and statistically significant, although its magnitude is smaller than that

Table X
Alternative Measures of Equity Issuance

This table reports results of analyses in which I use alternative measures of net equity issuance. In columns (1) and (2), the dependent variable is $SEO_{it} - REPO_{it}^{SDC}$. In columns (3) and (4), the dependent variable is $SEO_{it} - REPO_{it}^{Cpst}$. In columns (5) and (6), the dependent variable is $SEOD_{it}$. In columns (2), (4), and (6), I report results of analyses in which I exclude firms that never engage in SEOs during the sample period according to SDC. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	$SEO_{it} - REPO_{it}^{SDC}$		$SEO_{it} - REPO_{it}^{Cpst}$		$SEOD_{it}$	
	All (1)	SDC (2)	All (3)	SDC (4)	All (5)	SDC (6)
ΔPO_{it}	-0.034** (0.016)	-0.089** (0.040)	-0.048*** (0.017)	-0.101** (0.044)	-0.152** (0.077)	-0.435** (0.207)
$Tangibility_{it-1}$	-0.023* (0.014)	-0.053* (0.031)	-0.020 (0.015)	-0.058* (0.033)	-0.116* (0.068)	-0.288* (0.154)
$Investment_{it-1}$	-0.001 (0.001)	0.002 (0.002)	0.000 (0.001)	0.001 (0.003)	0.006 (0.007)	0.019 (0.013)
$Cash_{it-1}$	-0.029*** (0.008)	-0.072*** (0.016)	-0.045*** (0.009)	-0.077*** (0.018)	-0.160*** (0.038)	-0.352*** (0.077)
$Leverage_{it-1}$	0.001 (0.002)	0.003 (0.004)	0.005* (0.002)	0.003 (0.005)	0.012 (0.009)	0.020 (0.019)
$Salesgrowth_{it-1}$	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.002)	-0.002 (0.005)	-0.000 (0.007)
$Size_{it-1}$	-0.015*** (0.002)	-0.030*** (0.004)	-0.017*** (0.002)	-0.033*** (0.005)	-0.049*** (0.010)	-0.105*** (0.021)
$Tobins' Q_{it-1}$	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	-0.001 (0.001)	0.009*** (0.003)	0.014** (0.006)
$Cashflow_{it-1}$	0.003 (0.005)	0.001 (0.009)	-0.002 (0.005)	0.003 (0.010)	-0.005 (0.024)	-0.000 (0.045)
$BHAR_{it}$	0.009*** (0.002)	0.015*** (0.003)	0.010*** (0.002)	0.017*** (0.004)	0.033*** (0.007)	0.056*** (0.013)
$BHAR_{it-1}$	0.003*** (0.001)	0.004*** (0.002)	0.002** (0.001)	0.005*** (0.002)	0.008 (0.005)	0.014 (0.009)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	10,402	3,875	10,402	3,875	10,404	3,875
R^2	0.319	0.354	0.391	0.354	0.323	0.270

reported in Table IV. Because only 6% of firm-year observations are associated with SEOs and a number of sample firms never appear in the SDC database,¹⁵ I next repeat the analyses after excluding firms that report no SEOs during the

¹⁵ The low frequency of SEOs that I find is consistent with results reported by Lee and Masulis (2009). After excluding cases with insufficient financial data from CRSP and Compustat, their SEO sample consists of 963 events over the 1990 to 2002 period among U.S. issuers of common stocks listed on NYSE, Amex, or Nasdaq. This indicates that only 74 firms in the database engage

sample period. The results are presented in column (2). The magnitude of the coefficient on ΔPO_t increases significantly. The coefficient estimate suggests that a one-standard-deviation decrease in ΔPO_t is associated with an increase in $SEO_{minus}REPO_t^{SDC}$ of approximately 0.8% of total securities outstanding, which accounts for approximately 10.4% of the sample standard deviation of $SEO_{minus}REPO_t^{SDC}$. In columns (3) and (4), I report results of the regressions of $SEO_{minus}REPO_t^{Cpst}$. The magnitudes of the coefficients are slightly larger than those presented in columns (1) and (2). The coefficient estimates shown in column (4) suggest that a one-standard-deviation decrease in ΔPO_t is associated with an increase in $SEO_{minus}REPO_t^{Cpst}$ of approximately 0.9% of total securities outstanding, which accounts for approximately 11.0% of the sample standard deviation of $SEO_{minus}REPO_t^{SDC}$. In columns (5) and (6) I report results of the regressions of $SEOD_t$. When I include all sample firms in the analysis, the coefficient on ΔPO_t is -0.152 and statistically significant. When I exclude firms that never engage in SEOs during the sample period according to SDC, the coefficient on ΔPO_t is -0.435 and statistically significant. Overall, the results in this table indicate that the negative association between ΔPO_t and contemporaneous equity issuance remains strong for measures of equity issuance that are not affected by inactive equity activities and thus suggest that inactive equity activities are not driving the negative association between $NetEquityIssue_t$ and ΔPO_t .

VI. Do Managers Believe that ΔPO_t Predicts Mispricing?

In this section, I examine whether managers appear to believe that ΔPO_t predicts mispricing. To do so, I first examine managers' private trading activities. If managers believe that ΔPO_t predicts mispricing, they will sell their firms' shares when ΔPO_t predicts overvaluation and purchase shares when ΔPO_t predicts undervaluation. In addition, I analyze the association between ΔPO_t and contemporaneous option exercises on the part of managers. Because, according to Ofek and Yermack (2000), nearly all executive stock option exercises are followed by share sales, managers will exercise options when ΔPO_t predicts overvaluation.

The results of the analyses of managers' trading activities are reported in Table XI. In Panel A, I conduct the analyses at the manager level. In these analyses, I construct a panel of managers for each firm. In particular, I define the first (last) year a given manager reports sales or purchases of equity as the first (last) year the manager works for the firm. If no trading is reported in a given year, I set the number of shares sold and the number of shares purchased to zero. In odd-numbered columns I control for $BHAR_t$ and $BHAR_{t-1}$ as well as for manager and year fixed effects. In even-numbered columns I additionally control for firm characteristics. For columns (1) and (2) the dependent variable is $NetSell_t$. The coefficient on ΔPO_t is negative and statistically significant at

in SEOs each year. Among the firms that engage in SEOs at least once during the sample period, 65% issued only once and 25% issued only twice during the 13-year period.

Table XI
Insider Trading

This table reports results on managers' trading activities. In Panel A I report results at the manager level. In these analyses, I construct a panel of managers for each firm. In particular, I define the first (last) year a given manager reports sales or purchases of equity as the first (last) year the manager works for the firm. The dependent variables are $NetSell_t$, $Sell_t$, Buy_t , and $Sell/(Sell + Buy)_t$ in columns (1) and (2), (3) and (4), (5) and (6), and (7) and (8), respectively. Panel B reports results on trading activities executed by managers in aggregate at the firm level. The dependent variables are $NetSell_t^{Firm}$, $Sell_t^{Firm}$, Buy_t^{Firm} , and $Sell/(Sell + Buy)_t^{Firm}$ in columns (1) and (2), (3) and (4), (5) and (6), and (7) and (8), respectively. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the manager (firm) level in Panel A (B) and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Panel A. Manager level							
	$NetSell_t$		$Sell_t$		Buy_t		$Sell/(Sell + Buy)_t$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	-0.122*** (0.039)	-0.104*** (0.039)	-0.113*** (0.039)	-0.095** (0.039)	0.012*** (0.005)	0.011** (0.005)	-0.076*** (0.027)	-0.068** (0.027)
$Tangibility_{t-1}$		-0.486*** (0.096)		-0.506*** (0.095)		-0.015 (0.013)		-0.155** (0.073)
$Investment_{t-1}$		0.023* (0.013)		0.027** (0.013)		0.004** (0.002)		0.013* (0.007)
$Cash_{t-1}$		-0.065 (0.056)		-0.081 (0.055)		-0.013* (0.007)		0.039 (0.041)
$Leverage_{t-1}$		0.011 (0.013)		0.016 (0.012)		0.006** (0.002)		-0.044*** (0.016)
$Salesgrowth_{t-1}$		0.006 (0.006)		0.006 (0.006)		-0.001 (0.001)		0.002 (0.007)
$Size_{t-1}$		-0.034** (0.015)		-0.040*** (0.015)		-0.006*** (0.002)		0.026** (0.012)
$Tobins' Q_{t-1}$		0.046*** (0.006)		0.044*** (0.006)		-0.003*** (0.001)		0.025*** (0.004)
$Cash\ flow_{t-1}$		0.172*** (0.038)		0.169*** (0.037)		-0.005 (0.006)		-0.025 (0.039)

(Continued)

Table XI—Continued

	NetSell _{<i>t</i>}		Sell _{<i>t</i>}		Buy _{<i>t</i>}		Sell/(Sell + Buy) _{<i>t</i>}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
BHAR _{<i>t</i>}	0.082*** (0.008)	0.095*** (0.009)	0.082*** (0.008)	0.094*** (0.009)	0.000 (0.001)	−0.001 (0.001)	0.030*** (0.006)	0.044*** (0.007)
BHAR _{<i>t</i>−1}	0.103*** (0.008)	0.081*** (0.008)	0.099*** (0.008)	0.079*** (0.008)	−0.003*** (0.001)	−0.002* (0.001)	0.038*** (0.005)	0.031*** (0.005)
Manager FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
N	59,006	59,006	59,006	59,006	59,006	59,006	12,228	12,228
R ²	0.344	0.347	0.348	0.351	0.282	0.283	0.686	0.690

(Continued)

Table XI—Continued

	Panel B. Firm level							
	<i>NetSell</i> _{<i>t</i>} ^{<i>Firm</i>}				<i>Buy</i> _{<i>t</i>} ^{<i>Firm</i>}			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPO_t	-0.671** (0.286)	-0.604** (0.283)	-0.615** (0.284)	-0.550* (0.281)	0.075** (0.034)	0.072** (0.033)	-0.106** (0.047)	-0.100** (0.046)
<i>Tangibility</i> _{<i>t</i>-1}	-2.750***		-2.862***		-0.082		-0.078	
<i>Investment</i> _{<i>t</i>-1}		(0.663)		(0.663)		(0.084)		(0.104)
<i>Cash</i> _{<i>t</i>-1}		0.120 (0.076)		0.134* (0.076)		0.016* (0.010)		0.009 (0.009)
<i>Leverage</i> _{<i>t</i>-1}		-0.390 (0.378)		-0.481 (0.374)		-0.069 (0.050)		0.113** (0.055)
<i>Salesgrowth</i> _{<i>t</i>-1}		0.015 (0.070)		0.042 (0.068)		0.028** (0.013)		-0.044** (0.018)
<i>Size</i> _{<i>t</i>-1}		0.037 (0.045)		0.035 (0.044)		-0.002 (0.006)		0.004 (0.010)
<i>Tobins' Q</i> _{<i>t</i>-1}		-0.248** (0.111)		-0.280** (0.109)		-0.032*** (0.011)		0.071*** (0.016)
<i>Cashflow</i> _{<i>t</i>-1}		0.247*** (0.041)		0.234*** (0.040)		-0.016*** (0.004)		0.029*** (0.005)
<i>BHAR</i> _{<i>t</i>}		0.939*** (0.222)		0.925*** (0.214)		-0.024 (0.034)		0.102** (0.048)
<i>BHAR</i> _{<i>t</i>-1}	0.442*** (0.066)	0.504*** (0.073)	0.444*** (0.066)	0.498*** (0.072)	0.002 (0.007)	-0.006 (0.007)	0.039*** (0.010)	0.060*** (0.012)
	0.554*** (0.059)	0.434*** (0.053)	0.538*** (0.058)	0.421*** (0.053)	-0.016*** (0.005)	-0.011** (0.005)	0.056*** (0.008)	0.047*** (0.008)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	9,687	9,687	9,687	9,687	9,687	9,687	6,177	6,177
<i>R</i> ²	0.386	0.398	0.383	0.395	0.316	0.320	0.555	0.566

the 1% level in both columns, suggesting that managers engage in significantly more stock sales (purchases) when predisclosed POs predict overvaluation (undervaluation) of shares. The coefficient estimate in column (1) indicates that, for an average manager in the sample, a one-standard-deviation decrease in ΔPO_t is associated with an increase in net share sales of approximately 4.6%.

I next investigate whether the results in columns (1) and (2) are due to sales or purchases of shares. To this end, I run the regressions after separating sales and purchases of shares. In columns (3) and (4) the dependent variable is $Sell_t$. The coefficient on ΔPO_t is negative and statistically significant. In addition, the magnitudes of the coefficient estimates are similar to those in columns (1) and (2). These results suggest that the effects of ΔPO_t on $NetSell_t$ may be driven mostly by sales of shares. The coefficient estimate in column (3) indicates that, for an average manager in the sample, a one-standard-deviation decrease in ΔPO_t is associated with an increase in share sales of approximately 4.0%. In columns (5) and (6) the dependent variable is Buy_t . Although the magnitudes of the coefficients on ΔPO_t are approximately one-tenth the size of those in columns (3) and (4), they are as statistically and economically significant as those in columns (3) and (4). For example, the coefficient estimate in column (5) indicates that, for an average manager in the sample, a one-standard-deviation increase in ΔPO_t is associated with an increase in share purchases of approximately 6.2%. The small coefficients are consistent with the fact that transactions executed by managers are predominantly sales. In columns (7) and (8) the dependent variable is $Sell/(Sell + Buy)_t$. The coefficient on ΔPO_t is negative and statistically significant in both columns. The pattern of managers' trading activities therefore suggests that managers believe that ΔPO_t predicts mispricing. The coefficients on $BHAR_t$ and $BHAR_{t-1}$ are positive and statistically significant when the dependent variable is $NetSell_t$, $Sell_t$, or $Sell/(Sell + Buy)_t$. This result is consistent with prior studies. For example, Seyhun (1992) shows that insiders are more likely to sell shares following periods of price appreciation, and Piotroski and Roulstone (2005) find that insider trading is inversely related to recent returns.

I next analyze managers' trading activities in aggregate at the firm level. The results are presented in Panel B. The dependent variables are $NetSell_t^{Firm}$, $Sell_t^{Firm}$, Buy_t^{Firm} , and $Sell/(Sell + Buy)_t^{Firm}$ in columns (1) and (2), (3) and (4), (5) and (6), and (7) and (8), respectively. The interpretation of the coefficients on ΔPO_t is similar to that in Panel A in all columns. Overall, the results reported in this table are consistent with the view that managers believe that ΔPO_t predicts mispricing and use such information in private trading.

In Table XII, I report results of option-exercise analyses. In columns (1) and (2), I report results on the option exercises of CEOs. In columns (3) and (4), I report results on all executives' option exercises. In columns (1) to (4), the dependent variable is $OptExer_t$. In these tests, I control for manager fixed effects and standard errors are clustered at the manager level. The coefficient estimate on ΔPO_t is consistently negative and statistically significant in columns (1) to (4). For example, the coefficient estimate in column (1) indicates that a one-standard-deviation decrease in ΔPO_t is associated with an increase in

Table XII
Option Exercises

This table reports results on managers' option exercises. The dependent variable in columns (1) to (4), $OptExer_t$, is the number of shares acquired through an option exercise (Execucomp item `opt_exer_num`), divided by the sum of the number of shares acquired through an option exercise and unexercised exercisable options (Execucomp item `opt_unex_exer_num`). Columns (1) and (2) report results for option exercises on the part of CEOs. Columns (3) and (4) report results for option exercises on the part of all executives in Execucomp. Columns (5) and (6) report results of firm-level analyses, in which the dependent variable is $OptExer_t^{Firm}$. All variables are defined in Appendix A and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the manager level in columns (1) to (4) and at the firm level in columns (5) and (6). Standard errors are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	$OptExer_t$				$OptExer_t^{Firm}$	
	CEOs		All Executives		(5)	(6)
	(1)	(2)	(3)	(4)		
ΔPO_t	-0.216** (0.086)	-0.189** (0.086)	-0.228*** (0.081)	-0.192** (0.081)	-0.221** (0.086)	-0.189** (0.086)
$Tangibility_{t-1}$		-0.349*** (0.087)		-0.408*** (0.112)		-0.349*** (0.087)
$Investment_{t-1}$		-0.007 (0.008)		-0.006 (0.007)		-0.007 (0.008)
$Cash_{t-1}$		0.060 (0.037)		0.061 (0.037)		0.060 (0.037)
$Leverage_{t-1}$		-0.005 (0.009)		-0.011 (0.009)		-0.005 (0.009)
$Salesgrowth_{t-1}$		0.003 (0.009)		-0.006 (0.008)		0.003 (0.009)
$Size_{t-1}$		0.042*** (0.010)		0.052*** (0.011)		0.045*** (0.010)
$Tobins' Q_{t-1}$		0.023*** (0.004)		0.024*** (0.005)		0.023*** (0.004)
$Cashflow_{t-1}$		0.146*** (0.038)		0.126*** (0.043)		0.146*** (0.038)
$BHAR_t$	0.028*** (0.009)	0.050*** (0.009)	0.061*** (0.008)	0.073*** (0.009)	0.037*** (0.008)	0.050*** (0.009)
$BHAR_{t-1}$	0.045*** (0.007)	0.045*** (0.007)	0.073*** (0.008)	0.065*** (0.008)	0.053*** (0.007)	0.045*** (0.007)
Firm FE	No	No	No	No	Yes	Yes
Manager FE	Yes	Yes	Yes	Yes	No	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5,703	5,703	27,349	27,349	5,703	5,703
R^2	0.321	0.314	0.2231	0.2349	0.289	0.314

CEOs' $OptExer_t$ of approximately 1.9 percentage points, which accounts for 8.1% of the standard deviation and 14.3% of the mean value of CEOs' $OptExer_t$. The coefficients on $BHAR_t$ and $BHAR_{t-1}$ are positive and statistically significant in all columns, suggesting that option exercises are active following periods of price appreciation. In columns (5) and (6), I report results on option

exercises at the firm level. The dependent variable is $OptExer_t^{Firm}$. In these columns, I control for firm fixed effects and standard errors are clustered at the firm level. The coefficient on ΔPO_t is negative and statistically significant in both columns, suggesting that in aggregate managers engage in more option exercises when ΔPO_t predicts overvaluation of firm shares. Overall, the results in this table are consistent with managers believing that ΔPO_t predicts mispricing.

VII. Summary and Conclusions

This paper provides empirical evidence on the effects of information asymmetry-driven mispricing on security issuance. By exploiting changes in predisclosure POs as a proxy for information asymmetry-driven mispricing, I show that firms avoid (prefer) issuing equity and debt when they are undervalued (overvalued). However, the effects of information asymmetry-driven mispricing are significantly stronger for net equity issuance than net debt issuance. Consequently, undervaluation leads to increased leverage. In line with the interpretation that these results are due to information asymmetry problems, the effects are stronger for firms, periods, and securities associated with high information asymmetry. Managers' private trading and option exercises suggest that managers believe that changes in POs predict mispricing.

This paper contributes to studies on market timing. Unlike most studies on market timing, I capture mispricing by exploiting managerial information that is unknown to the public. This renders the results less open to alternative explanations. In addition, the measure makes it possible to distinguish information asymmetry-driven mispricing from investor irrationality-driven mispricing. This paper is also related to studies on the effects of information asymmetry on capital structure. Unlike most such studies, however, I consider the sign of mispricing caused by information asymmetry in addition to the magnitude of information asymmetry. I show that the sign of mispricing plays an important role in security issuance decisions. More importantly, the results in this paper suggest that disclosing POs and other variables associated with managers' perceptions of future firm performance on annual and quarterly filings would help mitigate information asymmetry problems between insiders and outsiders.

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Appendix A: Variable Definitions

Variables	
$Marketcap_t$	$CSHO_t \times PRCC_F_t$.
$Debt_t$	Total debt ($DLC_t + DLTt_t$).
PO_t	Total POs outstanding at the end of year t scaled by total assets (AT_t).
PO_t^{t+1}	The portion of PO_t that is due in year $t+1$ scaled by total assets (AT_t).
PO_t^{rest}	$PO_t - PO_t^{t+1}$.
ΔPO_t	$PO_t - PO_{t-1}$.
ΔPO_{next_t}	$PO_t^{t+1} - PO_{t-1}^t$.
ΔPO_{rest_t}	$PO_t^{rest} - PO_{t-1}^{rest}$.
$NetEquityIssue_t$	Issuance of common and preferred stock ($SSTK_t$) – Purchase of common and preferred stock ($PRSTKC_t$)/($Marketcap_{t-1} + Debt_{t-1}$).
$NetDebtIssue_t$	(Long-term debt issuance ($DLTIS_t$) – Long-term debt reduction ($DLTR_t$))/($Marketcap_{t-1} + Debt_{t-1}$).
$EquityIssueD_t$	An indicator variable that is equal to one if a firm issues equity valued at more than 1% of the lagged value of total securities, and zero otherwise.
$RepurchaseD_t$	An indicator variable that is equal to one if a firm repurchases equity valued at more than 1% of the lagged value of total securities, and zero otherwise.
$DebtIssueD_t$	An indicator variable that is equal to one if a firm issues debt valued at more than 1% of the lagged value of total securities, and zero otherwise.
$DebtReductionD_t$	An indicator variable that is equal to one if a firm reduces debt valued at more than 1% of the lagged value of total securities, and zero otherwise.
$Leverage_t$	$Debt_t / (Debt_t + \text{common equity } (CEQ_t))$.
$Investment_t$	Capital expenditures ($CAPX_t$)/property, plant, and equipment ($PPENT_{t-1}$).
$Tobin'sQ_t$	(Total assets (AT_t) – common equity (CEQ_t) + $Marketcap_t$)/total assets (AT_t).
$Size_t$	Natural logarithm of total assets (AT_t).
$Sales\ growth_t$	$(Sales_t - Sales_{t-1}) / Sales_{t-1}$.
$Cash\ flow_t$	Earnings before interest, taxes, depreciation and amortization ($EBITDA_t$)/total assets (AT_{t-1}).
$Cash_t$	Cash and cash equivalents (CHE_t)/total assets (AT_t).
$Tangibility_t$	Property, plant, and equipment ($PPENT_t$)/total assets (AT_t).
$Analyst\ coverage_t$	The number of analysts following the firm in the I/B/E/S database.
$R\&D_t$	R&D (XRD_t)/sales ($SALE_t$).
Age_t	The number of years since a firm first appears in the Compustat database.
$BHAR(-1,1)$	Size and BM-adjusted buy-and-hold abnormal returns during the three-day window starting one day before the 10-K filing. I use size and BM-matched portfolios as benchmarks to compute abnormal returns. In constructing benchmark portfolios, I first sort the firms into quintiles based on the 20th, 40th, 60th, and 80th percentiles of the market value of shareholders' equity. I then sort each size quintile portfolio into five equal BM quintiles.

$BHAR_t$	Size and BM-adjusted buy-and-hold abnormal returns during fiscal year t . In particular, I add daily abnormal returns of all trading days during t .
ES_t	Actual earnings minus the last individual analyst forecast, scaled by price.
$SEO_{minus}REPO_t^{SDC}$	$(SEO_t - Repurchases_t)/(Marketcap_{t-1} + Debt_{t-1})$, where SEO_t and $Repurchases_t$ are acquired from the SDC database, scaled by lagged total securities outstanding.
$SEO_{minus}REPO_t^{Cpst}$	$(SEO_t - \text{Purchase of common and preferred stock (PRSTK}_t))/(\text{Marketcap}_{t-1} + Debt_{t-1})$, where SEO_t is obtained from the SDC database.
$SEOD$	An indicator variable that is equal to one if a firm engages in SEOs in a given year and zero otherwise.
$PublicDebtIssue_t$	Bonds and notes minus lagged bonds and notes, scaled by the lagged value of total securities, where bonds and notes are computed as the sum of senior bonds and notes and subordinated bonds and notes.
$PrivateDebtIssue_t$	Nonpublic debt minus lagged nonpublic debt, scaled by the lagged value of total securities.
DEF_t	Cash dividends plus investment plus the change in working capital minus internal cash flow, scaled by lagged assets.
$NetSell_t$	The number of shares sold minus the number of shares purchased, times 1,000, scaled by the lagged total number of shares outstanding.
$Sell_t$	The number of shares sold times 1,000, scaled by the lagged total number of shares outstanding.
Buy_t	The number of shares purchased times 1,000, scaled by the lagged total number of shares outstanding.
$Sell/(Sell + Buy)_t$	The number of shares sold, over the sum of the number of shares purchased and the number of shares sold.
$NetSell_t^{Firm}$	The sum of $NetSell_t$ of all managers in a firm in a given year.
$Sell_t^{Firm}$	The sum of $Sell_t$ of all managers in a firm in a given year.
Buy_t^{Firm}	The sum of Buy_t of all managers in a firm in a given year.
$Sell/(Sell + Buy)_t^{Firm}$	The average value of $Sell/(Sell + Buy)_t$ across all managers within a firm in a given year.
$OptExer_t$	The number of shares acquired through an option exercise (Execucomp item opt_exer_num), divided by the sum of the number of shares acquired through an option exercise and unexercised exercisable options (opt_unex_exer_num).
$OptExer_t^{Firm}$	The average value of $OptExer_t$ across all managers within a firm in a given year.

Appendix B: Alternative Measures of ΔPO_t

A potential concern arising from the way ΔPO_t is defined is that changes in POs could have been expected by investors, based on available information from year $t-1$. For example, investors would expect ΔPO_t to be positive if firm characteristics as of $t-1$ predict that PO_t will be larger than PO_{t-1} . Although the results presented in Table III suggest that ΔPO_t is news to

Table BI
Alternative Measures of Changes in POs

Column (1) ((4)) of this table presents results of a regression of PO_t on PO_{t-1} (PO_{t-1} and PO_{t-2}) and control variables as of $t-1$. I estimate an expected value of PO_t ($E[PO_t]$) using the results shown in column (1) ((4)). I then define $\widehat{\Delta PO}_t$ as PO_t minus $E[PO_t]$. Results of regressions of $NetEquityIssue_t$ ($NetDebtIssue_t$) on $\widehat{\Delta PO}_t$ are presented in columns (2) and (3) ((5) and (6)), respectively. All variables are defined in the [Appendix](#) and are winsorized at 1% in both tails of the distribution. In each column, I report estimated coefficients from OLS regressions. Standard errors are clustered at the firm level and are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	PO_t	$NetEquity$ $Issue_t$	$NetDebt$ $Issue_t$	PO_t	$NetEquity$ $Issue_t$	$NetDebt$ $Issue_t$
	(1)	(2)	(3)	(4)	(5)	(6)
PO_{t-1}	0.282*** (0.036)			0.219*** (0.038)		
PO_{t-2}				0.042 (0.028)		
$\widehat{\Delta PO}_t$		-0.021* (0.012)	-0.012 (0.012)		-0.028** (0.013)	-0.019 (0.013)
$Tangibility_{t-1}$	-0.001 (0.017)	-0.024 (0.045)	0.054*** (0.017)	0.001 (0.018)	-0.041 (0.048)	0.076*** (0.019)
$Investment_{t-1}$	-0.000 (0.001)	0.002 (0.004)	0.001 (0.001)	0.000 (0.001)	0.000 (0.005)	0.002 (0.001)
$Cash_{t-1}$	0.009 (0.007)	-0.142*** (0.023)	0.003 (0.007)	0.007 (0.009)	-0.140*** (0.027)	0.009 (0.009)
$Leverage_{t-1}$	0.003 (0.002)	-0.011 (0.016)	-0.022*** (0.004)	0.004* (0.002)	-0.003 (0.015)	-0.021*** (0.005)
$Salesgrowth_{t-1}$	-0.000 (0.001)	-0.001 (0.003)	0.000 (0.001)	0.000 (0.001)	0.002 (0.004)	-0.000 (0.001)
$Size_{t-1}$	0.001 (0.002)	-0.062*** (0.008)	-0.017*** (0.002)	0.000 (0.003)	-0.062*** (0.009)	-0.019*** (0.002)
$Tobins'Q_{t-1}$	-0.000 (0.001)	-0.004** (0.002)	0.000 (0.001)	-0.001 (0.001)	-0.004* (0.002)	-0.000 (0.001)
$Cashflow_{t-1}$	-0.007 (0.007)	-0.025 (0.019)	0.015*** (0.005)	-0.014 (0.010)	-0.059* (0.031)	0.026*** (0.008)
$BHAR_{t-1}$	0.001 (0.001)	0.003 (0.002)	-0.001 (0.001)	0.001 (0.001)	0.004 (0.003)	-0.001 (0.001)
$BHAR_t$		0.021*** (0.005)	-0.007*** (0.001)		0.024*** (0.005)	-0.007*** (0.001)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	10,402	10,402	10,402	9,030	9,030	9,030
R^2	0.677	0.350	0.204	0.702	0.369	0.217

investors around 10-K issuance, I address this concern by using unexpected changes in POs. In particular, I estimate an expected value of PO_t ($E[PO_t]$) using POs of the previous year(s) and firm characteristics as of $t-1$. I then define $\widehat{\Delta PO}_t$ as PO_t minus $E[PO_t]$.

Table BI reports the results. In column (1) I present results of a regression of PO_t on PO_{t-1} and firm characteristics as of $t-1$. The only coefficient that is statistically and economically significant is PO_{t-1} , suggesting that other publicly available information does not predict PO_t . Nevertheless, I use the results presented in column (1) to predict $E[PO_t]$, compute $\widehat{\Delta PO}_t$, and run regressions of security issuance. The results of the security issuance regressions are reported in columns (2) and (3). The dependent variable is $NetEquityIssue_t$ in column (2). The coefficient estimate on $\widehat{\Delta PO}_t$ is negative and significant at the 10% level. The dependent variable is $NetDebtIssue_t$ in column (3). The coefficient on $\widehat{\Delta PO}_t$ is negative but statistically insignificant.

I next add PO_{t-2} to the information set for predicting PO_t . The results of the regression of PO_t are presented in column (4). The coefficient on PO_{t-1} remains similar to that in column (1). In contrast, the estimated coefficient on PO_{t-2} is neither statistically nor economically significant. The coefficients on other control variables generally remain similar to those shown in column (1). The results of the regression of $NetEquityIssue_t$ are reported in column (5). The economic and statistical significance of the coefficient on $\widehat{\Delta PO}_t$ increase slightly relative to those reported in column (2). In contrast, the coefficient on $\widehat{\Delta PO}_t$ is statistically insignificant, as reported in column (6), for which the dependent variable is $NetDebtIssue_t$.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.